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ANGULAR MOMENTUM TRANSFER FROM SWIFT ELECTRONS TO SPHERICAL NANOPARTICLES ACROSS THE ENTIRE NANOSCALE

PROTOCOLO DE INVESTIGACIÓN

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1 Introduction

1.1 Background

Since the mid-20th century, various techniques have been proposed for manipulating objects at the micro and nanoscale [1–7]. Optical tweezers, based on the electromagnetic forces produced by focused light beams, have been widely employed for trapping and manipulating microobjects, including viruses and bacteria, thereby significantly impacting technological development, particularly in the field of medicine [1–3].

In 2004, García de Abajo introduced the possibility of manipulating nanoobjects using transmission electron microscopes (TEMs) [8]. Subsequently, experimental evidence has shown that TEMs induce motion and rotation in nanoparticles (NPs) [9, 10], leading to the development of a manipulation technique termed *electron tweezers* [11–13], referencing optical tweezers.

In various experimental studies on electron tweezers, it has been observed that the transfer of angular and linear momentum from the electron beam to a nanoparticle (NP) depends on both the speed of the electrons in the beam and the impact parameter —the effective distance between the trajectory of the electron beam and the center of the NP [9–11, 13–15]. By modifying the impact parameter, it is possible to induce either attractive or repulsive interaction between the electron beam and the NP, and it is also feasible to alter the direction of the induced rotation on the NP [11, 13, 14].

The scanning transmission electron microscope (STEM) forms images by employing focused electron beams that scan the area of interest, interacting with the sample [11]. Figure 1.1 illustrates a pair of gold particles, one large and one small. The small NP is scanned by the beam of a STEM within the region outlined in white. The STEM electron beam scans line by line, and during scanning, the electron beam remains stationary 20% of the time at the beginning of each line, awaiting a synchronization signal, resulting in a net current of electrons traveling away from the NP. This current is depicted as a shaded blue region in Figure 1.1. Therefore, although the electron beam scans the entire area of interest, it can be considered an effective impact parameter with respect to the center of the NP.

Figure 1.2 presents results reported in Ref. [11] showing six STEM images of a gold NP with a diameter of ~ 1.5 nm in the presence of another NP with a diameter of ~ 5 nm, at different times. In all images of Figure 1.2, the effective impact parameter is located to the left of the NPs (near the left edge of the image). The top three images of Figure 1.2, taken with an effective impact parameter of ~ 4.5 nm, demonstrate

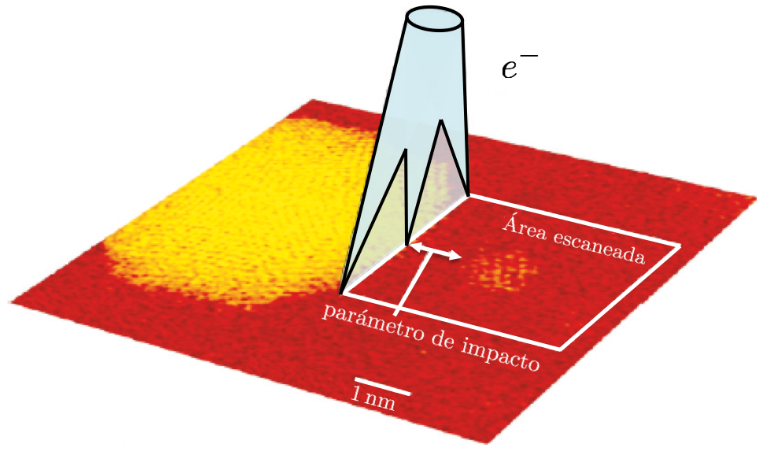

Figure 1.1: Schematics of the interaction within a STEM between an electron beam and a pair of gold nanoparticles, reproduced and adapted from Ref. [11].

an attractive interaction between the beam and the small NP (the large NP is used as a reference for displacement), as it can be observed that the NP approaches the left edge of the images, crossing the vertical white dashed line placed as a visual aid. Conversely, in the bottom images of Figure 1.2, the effective impact parameter is ~ 1 nm, and it is observed that the small NP moves away from the beam, crossing the white dashed line in the opposite direction, thus approaching the right edge of the image, indicating a repulsive interaction. Moreover, using the guide lines drawn on the facets of the large NP, forming a polygon, it can be appreciated that in the top three images, the large NP rotates clockwise, while in the bottom ones, upon changing the impact parameter, it rotates counterclockwise.

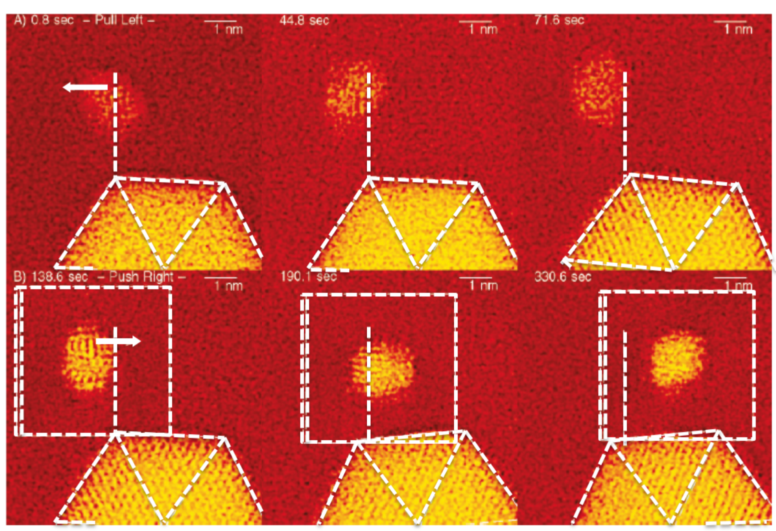

Figure 1.2: Results reproduced and adapted from Ref. [11] showing linear and angular momentum transfer from an electron beam in a STEM to gold nanoparticles, supported on an amorphous carbon substrate.

Although the momentum transfer from the electron beam to NPs is observed in the experiment, the development of electron tweezers technique require a theoretical understanding of the problem. Electron beams in a STEM can reach kinetic energies of 400 keV, with an electric current on the order of pA, equivalent to a pulse of fast electrons traveling at a constant speed of up to $0.83\,c$ (where c is the speed of light). From this, it is deduced that the emission time of each electron is $\sim 10^{-8}$ s. Since the lifetime of excitations within a metal is typically $\sim 10^{-14}$ s [16], it can be assumed that the NP interacts with one electron at a time [17–19]. Electron beams in STEM studies have been observed to deviate from a straight path by angles on the order of milliradians [19–21], so they can be considered to move in a straight line as long as the electrons travel outside the NP. Therefore, the trajectory of the fast electron can be modeled as $\vec{\mathbf{r}}(t) = (b, 0, vt)$, where v is the electron speed and b is the distance between the center of the NP and the electron trajectory (impact parameter), as shown in Figure 1.3. The electromagnetic response of the NP is modeled by its dielectric function $\epsilon(\omega)$.

From the perspective of quantum mechanics, an electron is considered to be a point-like particle with a fundamental electric charge. However, due to vacuum fluctuations, it has been estimated to have a finite size on the order of its Compton wavelength, which is approximately $\lambda_C = 2.424 \times 10^{-3}$ nm. In the case of a fast electron, such as those traveling within a STEM, its de Broglie wavelength is on the order of $\lambda_B \sim 10^{-2}$ nm [22]. Therefore, in this work, distances from the NP to the electron greater than λ_C and λ_B are considered. In a previous work [19], the interaction between electron beams and spherical NPs was addressed from the standpoint of quantum mechanics. Remarkably, it was concluded that for distances from the NP to the electron greater than λ_C and λ_B, a classical description of the problem is sufficient.

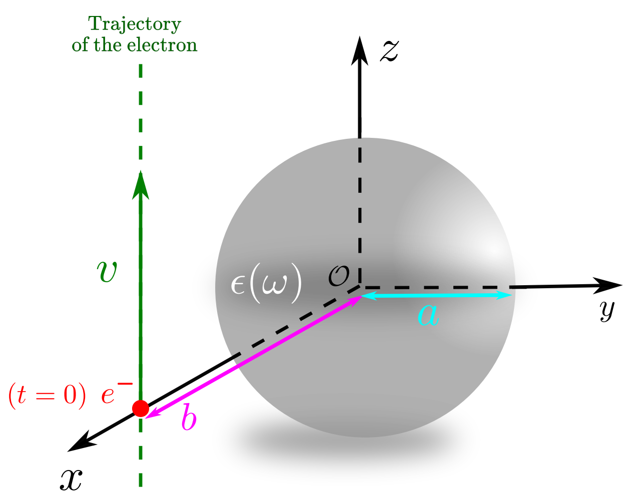

Figure 1.3: Nanoparticle with radius a, centered at the origin, modeled using a dielectric function $\epsilon(\omega)$, in vacuum. The trajectory of the electron (red dot), with impact parameter b and traveling at constant speed v, is shown as a dashed green line.

The interaction between electron beams and spherical NPs has also been studied from the perspective of classical electrodynamics in previous works [8, 15, 23–25]. The aforementioned studies have focused on calculating linear momentum transfer by solving Maxwell's equations in the frequency domain. To achieve this, a multipolar expansion has been employed to separate the electric and magnetic contributions to the interaction, as well as the contribution of each multipolar order. However, caution is necessary when choosing the dielectric function to model the electromagnetic response of NPs. Since the dielectric function is experimentally measured over a finite frequency range, it is necessary to extrapolate and interpolate it to perform the integration of the Maxwell stress tensor over the entire frequency space. Without sufficient care, this process may result in a non-causal dielectric function, meaning it does not satisfy the Kramers-Kronig relations. Although previous studies [8, 15, 23–25] successfully reproduced the repulsive behavior of the interaction, recent studies have shown that these works obtained non-physical results by modeling the electromagnetic response of the NP using non-causal dielectric functions [26, 27]. In these recent works, it is demonstrated that if the non-causal behavior of the dielectric functions is eliminated, the experimentally reported repulsive interaction does not appear. It is interesting to note that in Ref. [27], the integrals in frequency space are solved semi-analytically, which were previously solved numerically in Refs. [8, 15, 23–25]. This allows for an exact understanding of the contribution in frequency space of each multipole, electric or magnetic, to the linear momentum transfer, thereby enabling the calculation of linear momentum transfer from fast electrons to large NPs (up to $a = 50$ nm).

The electron tweezers technique will benefit from a detailed theoretical study of angular momentum transfer (AMT). Previous works have discussed angular dynamics briefly (see, for example, Ref. [18]), but recent studies have enabled its calculation in small NPs, up to $a = 5$ nm, using two different methods. The first method models the electromagnetic response of the NP as a point dipole $\vec{\mathbf{p}}$, using the polarizability tensor, which is valid only for small NPs [28]. The second method numerically solves the surface integrals of the Maxwell stress tensor [22, 29] in frequency space. However, numerical computation limits the size of NPs that can be studied (smaller than 5 nm, for realistic dielectric functions such as gold or bismuth), due to the required computation time, as reported in Refs. [22, 29]. Therefore, the methods developed previously only allow for the calculation of AMT in small nanoparticles, up to $a = 5$ nm.

It has been demonstrated that there are terms in the Maxwell stress tensor, corresponding to the external fields produced by the electron, that do not contribute to the total AMT [29]. Therefore, it has been shown that the integral of the Maxwell stress tensor containing only the electron's electromagnetic fields must vanish. The interaction term in the stress tensor, which includes both the electron's electromagnetic fields and the fields scattered by the nanoparticle, contributes the most to momentum transfer, and the term including only the fields scattered by the nanoparticle, although small, does not vanish [22, 27].

1.2 General objective and specific objectives

The general objective of this work is to develop a new methodology for the theoretical study of angular momentum transfer from a swift electron AMT to a spehrical NP, using a classical electrodynamics approach.

Among the key specific objectives, the first one is to derive an analytical solution for the integrals of the Maxwell stress tensor in frequency domain. Previously, these integrals were computed numerically using the cubature method.[1] The analytical solution will provide an efficient and systematic study of fast electron AMT to NPs with radii up to 50 nm, thus covering the entire nanoscale. In Section 2, the theoretical framework and methods necessary for deducing the electromagnetic fields generated by a relativistic electron in uniform rectilinear motion are developed. Additionally, the scattered electromagnetic fields by a NP centered at the origin, whose electromagnetic response is characterized by its dielectric function $\epsilon(\omega)$, are discussed. The AMT of the fast electron to the NP is then derived through an integral of the Maxwell stress tensor in frequency space, based on the principle of angular momentum conservation in electrodynamics.

The second specific objective is to apply the analytical solution to study a variety of materials, im-

[1] The quadrature method is different from the cubature method. Cubatures are used to numerically solve multidimensional integrals [30].

plementing a numerical code that computes the frequency-domain integral using quadrature methods. For further details of the code, the reader may visit Ref. [31]. In Section 3, the preliminary results are presented, showcasing the calculated solution to the Maxwell stress tensor integral in frequency space, thereby enabling efficient AMT computation.

Lastly, the final specific objective is to extend the AMT calculations across the entire nanoscale for different materials. To date, such calculations have primarily focused on Drude-like materials. This work represents the first time that the AMT has been calculated using dielectric functions of more realistic materials, marking a significant advancement in the field.

2
Theory, Methods & Techniques

In this Section, the classical electrodynamics approach is employed to describe the interaction of a fast electron traveling at a constant velocity in the vicinity of a spherical nanoparticle (NP). In particular, the angular momentum transfer (AMT) from the electron to the NP is analyzed. In a previous work by García de Abajo [19], it is justified that, under the conditions described in the Introduction Section, a quantum description of the phenomenon is not necessary.

During the mathematical development, equations will be presented in the International System of Units (SI) in **black**, and in (magenta and within parentheses)[1] the necessary factor to express the equation in the *cgs* system. For example, the force between two point charges q_1 and q_2 separated by a distance r will be written as

$$\vec{\mathbf{F}} = (4\pi\epsilon_0)\frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\hat{r}. \tag{2.1}$$

2.1 Conservation of Angular Momentum in Electrodynamics

The Maxwell equations are [32]

$$\begin{aligned} \nabla\cdot\vec{\mathbf{E}} &= (4\pi\epsilon_0)\frac{\rho_{\text{tot}}}{\epsilon_0}, & \nabla\times\vec{\mathbf{E}} &= -\left(\frac{1}{c}\right)\frac{\partial\vec{\mathbf{B}}}{\partial t}, \\ \nabla\cdot\vec{\mathbf{B}} &= 0, & \nabla\times\vec{\mathbf{B}} &= \left(\frac{4\pi}{\mu_0 c}\right)\mu_0\vec{\mathbf{J}}_{\text{tot}} + (c)\frac{1}{c^2}\frac{\partial\vec{\mathbf{E}}}{\partial t}, \end{aligned} \tag{2.2}$$

where $\vec{\mathbf{E}}$ is the electric field, $\vec{\mathbf{B}}$ is the magnetic field, ρ_{tot} is the total charge density, $\vec{\mathbf{J}}_{\text{tot}}$ is the total current density, c is the speed of light, ϵ_0 is the vacuum permittivity, and μ_0 is the vacuum permeability. In terms of the potentials ϕ and $\vec{\mathbf{A}}$, the Maxwell equations are given as follows [32]

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial}{\partial t}\right)\phi\left(\vec{\mathbf{r}},t\right) = -(4\pi\epsilon_0)\frac{\rho_{\text{tot}}\left(\vec{\mathbf{r}},t\right)}{\epsilon_0}, \tag{2.3}$$

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial}{\partial t}\right)\vec{\mathbf{A}}\left(\vec{\mathbf{r}},t\right) = -\left(\frac{4\pi}{\mu_0 c}\right)\mu_0\vec{\mathbf{J}}_{\text{tot}}\left(\vec{\mathbf{r}},t\right). \tag{2.4}$$

[1] The **International System** is used to express equations in their standard textbook form. However, expressions in the (cgs) system are ideal for a numerical description of the problem, which is proposed as future work.

During the theoretical development, the Lorenz gauge will be used, $\nabla \cdot \vec{\mathbf{A}} + (1/c)\partial_t \phi = 0$, so that the electromagnetic fields, in terms of the potentials, are written as

$$\vec{\mathbf{E}}\left(\vec{\mathbf{r}}, t\right) = -\nabla\phi\left(\vec{\mathbf{r}}, t\right) - \left(\frac{1}{c}\right)\frac{\partial}{\partial t}\vec{\mathbf{A}}\left(\vec{\mathbf{r}}, t\right), \tag{2.5}$$

$$\vec{\mathbf{B}}\left(\vec{\mathbf{r}}, t\right) = \nabla \times \vec{\mathbf{A}}\left(\vec{\mathbf{r}}, t\right). \tag{2.6}$$

The expression for the conservation of linear momentum in electrodynamics is [32]

$$\frac{\partial}{\partial t}\left[\vec{\mathbf{p}}^{\,\mathrm{mec}}\left(\vec{\mathbf{r}}, t\right) + \vec{\mathbf{p}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}}, t\right)\right] = \nabla \cdot \overleftrightarrow{\mathbf{T}}\left(\vec{\mathbf{r}}, t\right), \tag{2.7}$$

where $\vec{\mathbf{p}}^{\,\mathrm{mec}}\left(\vec{\mathbf{r}}, t\right)$ represents the volumetric density of mechanical linear momentum, and $\vec{\mathbf{p}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}}, t\right)$ represents the volumetric density of electromagnetic linear momentum

$$\vec{\mathbf{p}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}}, t\right) = \left(\frac{c}{4\pi}\right)\frac{1}{c^2}\vec{\mathbf{E}}\left(\vec{\mathbf{r}}, t\right) \times \vec{\mathbf{B}}\left(\vec{\mathbf{r}}, t\right), \tag{2.8}$$

and $\overleftrightarrow{\mathbf{T}}\left(\vec{\mathbf{r}}, t\right)$ is the Maxwell stress tensor given by [32]

$$T_{ij}\left(\vec{\mathbf{r}}, t\right) = \frac{\epsilon_0}{(4\pi\epsilon_0)}\left[E_i\left(\vec{\mathbf{r}}, t\right)E_j\left(\vec{\mathbf{r}}, t\right) - \frac{\delta_{ij}}{2}E^2\left(\vec{\mathbf{r}}, t\right)\right] + \frac{\mu_0}{(4\pi\mu_0)}\left[H_i\left(\vec{\mathbf{r}}, t\right)H_j\left(\vec{\mathbf{r}}, t\right) - \frac{\delta_{ij}}{2}H^2\left(\vec{\mathbf{r}}, t\right)\right], \tag{2.9}$$

where $T_{ij}\left(\vec{\mathbf{r}}, t\right)$ represents the ij entry of $\overleftrightarrow{\mathbf{T}}\left(\vec{\mathbf{r}}, t\right)$, $E_i\left(\vec{\mathbf{r}}, t\right)$ is the i-th component of the electric field $\vec{\mathbf{E}}\left(\vec{\mathbf{r}}, t\right)$, $H_i\left(\vec{\mathbf{r}}, t\right)$ is the i-th component of the field $\vec{\mathbf{H}}\left(\vec{\mathbf{r}}, t\right)$, and δ_{ij} is the Kronecker delta.

From Eq. (2.7), the conservation of angular momentum in electrodynamics can be deduced as follows

$$\frac{\partial}{\partial t}\left[\vec{\boldsymbol{\ell}}^{\,\mathrm{mec}}\left(\vec{\mathbf{r}}, t\right) + \vec{\boldsymbol{\ell}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}}, t\right)\right] = \vec{\mathbf{r}} \times \nabla \cdot \overleftrightarrow{\mathbf{T}}\left(\vec{\mathbf{r}}, t\right), \tag{2.10}$$

where $\vec{\boldsymbol{\ell}}^{\,\mathrm{mec}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}^{\,\mathrm{mec}}$ and $\vec{\boldsymbol{\ell}}^{\,\mathrm{em}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}^{\,\mathrm{em}}$ are the volumetric densities of mechanical and electromagnetic angular momentum, respectively.

If we define the tensor $\overleftrightarrow{\mathbf{M}}\left(\vec{\mathbf{r}}, t\right) = \vec{\mathbf{r}} \times \overleftrightarrow{\mathbf{T}}\left(\vec{\mathbf{r}}, t\right)$ —or using index notation and Einstein's summation convention $M_{jk}\left(\vec{\mathbf{r}}\, t\right) = \epsilon_j{}^{li}\, r_l\, T_{ik}\left(\vec{\mathbf{r}}\, t\right)$— and calculate the divergence of $\overleftrightarrow{\mathbf{M}}$, we obtain

$$\begin{aligned}
\left(\nabla \cdot \overleftrightarrow{\mathbf{M}}\right)_j &= \delta^{nk}\partial_n M_{jk} = \delta^{nk}\partial_n \epsilon_j{}^{li} r_l T_{ik} = \delta^{nk}\epsilon_j{}^{li}\partial_n r_l T_{ik}, \\
&= \delta^{nk}\epsilon_j{}^{li}\left(\delta_{nl}T_{ik} + r_l\partial_n T_{ik}\right) = \delta^{nk}\epsilon_j{}^{li} r_l \partial_n T_{ik}, \\
&= \delta^{nk}\epsilon_j{}^{li} r_l \partial_n T_{ik} = \epsilon_j{}^{li} r_l \partial^k T_{ik} = \epsilon_j{}^{li} r_l \left(\nabla \cdot \overleftrightarrow{\mathbf{T}}\right)_i,
\end{aligned} \tag{2.11}$$

so that

$$\left(\nabla \cdot \overleftrightarrow{\mathbf{M}}\right)_j = \left(\vec{\mathbf{r}} \times \nabla \cdot \overleftrightarrow{\mathbf{T}}\right)_j, \tag{2.12}$$

where it has been identified that $\delta^{nk}\delta_{nl}\,\epsilon_j{}^{li}T_{ik} = \epsilon_j{}^{ni}T_{in} = 0$, because the Maxwell stress tensor is symmetric ($T_{in} = T_{ni}$) and the Levi-Civita symbol is antisymmetric $\left(\epsilon_j{}^{ni} = -\epsilon_j{}^{in}\right)$.

From Eqs. (2.10) and (2.12), the conservation of angular momentum is written as

$$\boxed{\frac{\partial}{\partial t}\left[\vec{\boldsymbol{\ell}}^{\,\mathrm{mec}}\left(\vec{\mathbf{r}},t\right)+\vec{\boldsymbol{\ell}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}},t\right)\right]=\nabla\cdot\overleftrightarrow{\mathbf{M}}\left(\vec{\mathbf{r}},t\right),} \tag{2.13}$$

which is a local equation. To represent the global conservation of angular momentum, Eq. (2.13) is integrated over a volume V bounded by a surface S as follows

Global angular momentum transfer in classical electrodynamics

$$\frac{d}{dt}\left[\vec{\mathbf{L}}^{\mathrm{mec}}\left(t\right)+\vec{\mathbf{L}}^{\mathrm{em}}\left(t\right)\right]=\oint_S\overleftrightarrow{\mathbf{M}}\left(\vec{\mathbf{r}},t\right)\cdot d\vec{\mathbf{S}}, \tag{2.14}$$

where the divergence theorem has been used in the last equality, and

$$\vec{\mathbf{L}}^{\mathrm{mec}}\left(t\right)=\int_V\vec{\boldsymbol{\ell}}^{\,\mathrm{mec}}\left(\vec{\mathbf{r}},t\right)\,dV \qquad \text{and} \qquad \vec{\mathbf{L}}^{\mathrm{em}}\left(t\right)=\int_V\vec{\boldsymbol{\ell}}^{\,\mathrm{em}}\left(\vec{\mathbf{r}},t\right)\,dV. \tag{2.15}$$

It is important to mention that the integration surface S contains the NP but does not intersect the trajectory of the electron, as shown in Fig. 2.1.

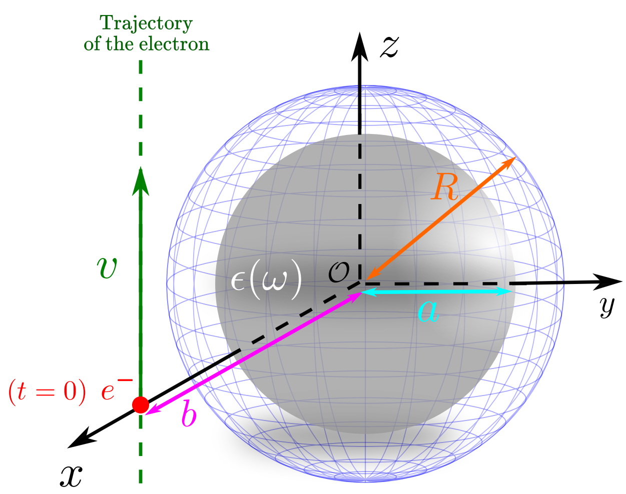

Figure 2.1: Spherical nanoparticle of radius a centered at the origin and characterized by a homogeneous dielectric response $\epsilon(\omega)$, enclosed by the integration surface S of radius R, along with the trajectory of the fast electron located at $\vec{\mathbf{r}}=(b,0,vt)$.

The system under investigation consists of a spherical nanoparticle (NP) of radius a, characterized by a homogeneous dielectric response $\epsilon(\omega)$ and centered at the origin, interacting with a fast electron whose trajectory is described by $\vec{\mathbf{r}}=(b,0,vt)$, as depicted in Fig. 2.1. To calculate the transfer of angular momentum (TMA) from the electron to the NP ($\Delta\vec{\mathbf{L}}$), Eq. (2.14) is integrated along the entire trajectory of the electron, or equivalently in time, as follows

$$\Delta\vec{\mathbf{L}}=\int_{-\infty}^{\infty}\frac{d}{dt}\vec{\mathbf{L}}^{\mathrm{mec}}\left(t\right)\,dt=\int_{-\infty}^{\infty}\oint_S\overleftrightarrow{\mathbf{M}}\left(\vec{\mathbf{r}},t\right)\cdot d\vec{\mathbf{S}}\,dt-\Delta\vec{\mathbf{L}}^{\mathrm{em}}, \tag{2.16}$$

where

$$\Delta\vec{\mathbf{L}}^{\text{em}} = \int_{-\infty}^{\infty} \frac{d}{dt}\vec{\mathbf{L}}^{\text{em}}\, dt = \vec{\mathbf{L}}^{\text{em}}(t\to\infty) - \vec{\mathbf{L}}^{\text{em}}(t\to-\infty). \tag{2.17}$$

The terms

$$\vec{\mathbf{L}}^{\text{em}}(t\to\pm\infty) = \epsilon_0\mu_0 \int_V \vec{\mathbf{r}} \times \left[\vec{\mathbf{E}}\,(t\to\pm\infty) \times \vec{\mathbf{H}}\,(t\to\pm\infty)\right] dV \tag{2.18}$$

are null because at time $t\to-\infty$ the electron is infinitely far away and has not interacted with the NP, meaning that the total electromagnetic fields $\{\vec{\mathbf{E}},\vec{\mathbf{H}}\}$, composed of the sum of external fields $\{\vec{\mathbf{E}}_{\text{ext}},\vec{\mathbf{H}}_{\text{ext}}\}$ and those scattered by the NP $\{\vec{\mathbf{E}}_{\text{scat}},\vec{\mathbf{H}}_{\text{scat}}\}$, are null within the integration volume V —$\vec{\mathbf{E}}(t\to-\infty)=\vec{\mathbf{0}}$ and $\vec{\mathbf{H}}(t\to-\infty)=\vec{\mathbf{0}}$—. On the other hand, for $t\to\infty$, the electron will also be infinitely far away, but it will have already interacted with the NP, resulting in the induction of charge and current distributions within the NP. However, these distributions will have vanished for $t\to\infty$ due to dissipative processes. Therefore, $\vec{\mathbf{L}}^{\text{em}}(t\to\pm\infty)=0$.

Then

$$\boxed{\Delta\vec{\mathbf{L}} = \int_{-\infty}^{\infty} \oint_S \overleftrightarrow{\mathbf{M}}\,(\vec{\mathbf{r}},t)\cdot d\vec{\mathbf{S}}\, dt,} \tag{2.19}$$

or using index notation

$$\Delta L_i = \int_{-\infty}^{\infty} \oint_S \epsilon_i{}^{lj} r_l T_{jk}\,(\vec{\mathbf{r}},t)\, n^k\, dS\, dt, \tag{2.20}$$

where n_i is the i-th component of the normal vector to the surface S.

For the calculation of angular momentum transfer, it is convenient to express the electromagnetic fields in the frequency domain. Using the following definition of the Fourier transform

$$\vec{\mathbf{F}}(\omega) = \int_{-\infty}^{\infty} \vec{\mathbf{F}}(t)\,\mathrm{e}^{\mathrm{i}\omega t}\, dt \qquad \text{y} \qquad \vec{\mathbf{F}}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \vec{\mathbf{F}}(\omega)\,\mathrm{e}^{-\mathrm{i}\omega t}\, d\omega, \tag{2.21}$$

where $\vec{\mathbf{F}}\in\{\vec{\mathbf{B}},\vec{\mathbf{H}}\}$, and for $\vec{\mathbf{F}}(t)$ to be a real function, it must hold in the frequency domain that $\vec{\mathbf{F}}(\omega)^* = \vec{\mathbf{F}}(-\omega)$, where $*$ denotes complex conjugation. To calculate the angular momentum transfer via Eq. (2.20), it is important to note that the time dependence is contained only in the Maxwell stress tensor $\overleftrightarrow{\mathbf{T}}\,(\vec{\mathbf{r}},t)$. Thus, the integral in time of the product of electric fields can be rewritten as follows

$$\begin{aligned}
\int_{-\infty}^{\infty} E_i\,(\vec{\mathbf{r}},t)\, E_j\,(\vec{\mathbf{r}},t)\; dt =& \int_{-\infty}^{\infty} \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} E_i\,(\vec{\mathbf{r}},\omega)\;\mathrm{e}^{-\mathrm{i}\omega t}\, d\omega\right] \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} E_j\left(\vec{\mathbf{r}},\omega'\right)\;\mathrm{e}^{-\mathrm{i}\omega' t}\, d\omega'\right] dt \\
=& \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}(\omega+\omega')t}\, dt\right] E_i\,(\vec{\mathbf{r}},\omega)\, E_j\left(\vec{\mathbf{r}},\omega'\right)\; d\omega\, d\omega' \\
=& \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \delta\left(\omega+\omega'\right) E_i\,(\vec{\mathbf{r}},\omega)\, E_j\left(\vec{\mathbf{r}},\omega'\right)\; d\omega\, d\omega' \\
=& \frac{1}{2\pi}\int_{-\infty}^{\infty} E_i\,(\vec{\mathbf{r}},\omega)\, E_j\,(\vec{\mathbf{r}},-\omega)\; d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} E_i\,(\vec{\mathbf{r}},\omega)\, E_j^*\,(\vec{\mathbf{r}},\omega)\; d\omega \\
=& \frac{1}{\pi}\int_0^{\infty} \mathrm{Re}\left[E_i\,(\vec{\mathbf{r}},\omega)\, E_j^*\,(\vec{\mathbf{r}},\omega)\right]\; d\omega.
\end{aligned} \tag{2.22}$$

For the components of the field $\vec{\mathbf{H}}$, an analogous process is followed.

Thus, Eq. (2.20) can be rewritten as

$$\boxed{\Delta L_i = \frac{1}{\pi}\int_0^\infty \oint_S \epsilon_i{}^{lj} r_l \mathcal{T}_{jk}\left(\vec{\mathbf{r}},\omega\right) n^k \, dS \, d\omega,} \tag{2.23}$$

where we have defined

$$\begin{aligned}\overleftrightarrow{\mathcal{T}}\left(\vec{\mathbf{r}},\omega\right) = \mathrm{Re}\Bigg[& \frac{\epsilon_0}{(4\pi\epsilon_0)}\vec{\mathbf{E}}\left(\vec{\mathbf{r}},\omega\right)\vec{\mathbf{E}}^*\left(\vec{\mathbf{r}},\omega\right) - \frac{\epsilon_0}{(4\pi\epsilon_0)}\frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}\left(\vec{\mathbf{r}},\omega\right)\cdot\vec{\mathbf{E}}_j^*\left(\vec{\mathbf{r}},\omega\right) \\ & + \frac{\mu_0}{(4\pi\mu_0)}\vec{\mathbf{H}}_i\left(\vec{\mathbf{r}},\omega\right)\vec{\mathbf{H}}_j^*\left(\vec{\mathbf{r}},\omega\right) - \frac{\mu_0}{(4\pi\mu_0)}\frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{H}}\left(\vec{\mathbf{r}},\omega\right)\cdot\vec{\mathbf{H}}^*\left(\vec{\mathbf{r}},\omega\right)\Bigg],\end{aligned} \tag{2.24}$$

where $\overleftrightarrow{\mathbf{I}}$ is the rank-2 identity tensor. Thus, the *spectral density* of angular momentum is defined as

$$\boxed{\mathcal{L}_i\left(\omega\right) = \frac{1}{\pi}\oint_S \epsilon_i{}^{lj} r_l \mathcal{T}_{jk}\left(\vec{\mathbf{r}},\omega\right) n^k \, dS,} \tag{2.25}$$

so that the angular momentum transfer is calculated as

$$\Delta\vec{\mathbf{L}} = \int_0^\infty \vec{\mathcal{L}}\left(\omega\right)\, d\omega. \tag{2.26}$$

In Cartesian coordinates, Eq. (2.25) can be written as

$$\mathcal{L}_x = \frac{1}{\pi}\left[\oint_S y\,\mathcal{T}_{zk}\, dS^k - \oint_S z\,\mathcal{T}_{yk}\, dS^k\right], \tag{2.27}$$

$$\mathcal{L}_y = \frac{1}{\pi}\left[\oint_S z\,\mathcal{T}_{xk}\, dS^k - \oint_S x\,\mathcal{T}_{zk}\, dS^k\right], \tag{2.28}$$

$$\mathcal{L}_z = \frac{1}{\pi}\left[\oint_S x\,\mathcal{T}_{yk}\, dS^k - \oint_S y\,\mathcal{T}_{xk}\, dS^k\right]. \tag{2.29}$$

Due to the problem of interaction between a fast electron and a spherical NP possessing reflection symmetry with respect to the xz plane, there cannot exist a component of transferred linear momentum in the y direction [23]. From this, it can be concluded that the angular momentum transferred to the NP cannot have x nor z components [22]. Therefore, it follows that only the spectral contribution $\mathcal{L}_y$ is nonzero. In terms of the spherical basis, the spectral components are given by the following expressions [22]

Nonvanishing spectral density component

$$\mathcal{L}_y = \frac{R}{\pi}\oint \left[\cos\varphi\,\mathcal{T}_{\theta r} - \cos\theta\sin\varphi\,\mathcal{T}_{\varphi r}\right] dS_r, \tag{2.30}$$

and

Vanishing spectral density components

$$\mathcal{L}_x = \frac{R}{\pi}\oint\left[\sin\varphi\,\mathcal{T}_{\theta r} + \cos\theta\cos\varphi\,\mathcal{T}_{\varphi r}\right] dS_r, \tag{2.31}$$

$$\mathcal{L}_z = \frac{R}{\pi}\oint \sin\theta\,\mathcal{T}_{\varphi r}\, dS_r, \tag{2.32}$$

where a spherical integration surface has been assumed, as shown in Fig. 2.1.

It is possible to separate the electric and magnetic contributions in the spectral angular momentum density, Eq. (2.26), so that the tensor can be separated as follows

$$\overleftrightarrow{\mathcal{T}}(\vec{\mathbf{r}},\omega) = \overleftrightarrow{\mathcal{T}}^{\mathrm{E}}(\vec{\mathbf{r}},\omega) + \overleftrightarrow{\mathcal{T}}^{\mathrm{H}}(\vec{\mathbf{r}},\omega), \tag{2.33}$$

where

$$\overleftrightarrow{\mathcal{T}}^{\mathrm{E}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\vec{\mathbf{E}}(\vec{\mathbf{r}},\omega)\,\vec{\mathbf{E}}^*(\vec{\mathbf{r}},\omega) - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}(\vec{\mathbf{r}},\omega)\cdot\vec{\mathbf{E}}^*(\vec{\mathbf{r}},\omega)\right] \tag{2.34}$$

and

$$\overleftrightarrow{\mathcal{T}}^{\mathrm{H}} = \frac{\mu_0}{(4\pi\mu_0)}\mathrm{Re}\left[\vec{\mathbf{H}}(\vec{\mathbf{r}},\omega)\,\vec{\mathbf{H}}^*(\vec{\mathbf{r}},\omega) - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{H}}(\vec{\mathbf{r}},\omega)\cdot\vec{\mathbf{H}}^*(\vec{\mathbf{r}},\omega)\right]. \tag{2.35}$$

Since we are considering a spherical integration surface S of radius R and $\hat{r}_i$ as the i-th component of the radial unit vector, the spectral angular momentum density $\mathcal{L}_i(\omega)$ can be expressed as follows

$$\mathcal{L}_i(\omega) = \frac{R^2}{\pi}\int_0^{4\pi}\left[\epsilon_i{}^{lj} r_l \mathcal{T}^{\mathrm{E}}_{jk}(\vec{\mathbf{r}},\omega)\, n^k + \epsilon_i{}^{lj} r_l \mathcal{T}^{\mathrm{H}}_{jk}(\vec{\mathbf{r}},\omega)\, n^k\right] d\Omega, \tag{2.36}$$

where only the solid angle Ω remains to be integrated. In Eq. (2.36) it is noticeable that the electric contribution is separated from the magnetic, allowing us to define both contributions as

$$\mathcal{L}^{\mathrm{E}}_i(\omega) = \frac{R^2}{\pi}\int_0^{4\pi}\left[\epsilon_i{}^{lj} r_l \mathcal{T}^{\mathrm{E}}_{jk}(\vec{\mathbf{r}},\omega)\, n^k\right] d\Omega, \tag{2.37}$$

$$\mathcal{L}^{\mathrm{H}}_i(\omega) = \frac{R^2}{\pi}\int_0^{4\pi}\left[\epsilon_i{}^{lj} r_l \mathcal{T}^{\mathrm{H}}_{jk}(\vec{\mathbf{r}},\omega)\, n^k\right] d\Omega. \tag{2.38}$$

Since we can separate the electromagnetic fields $\mathbf{E}$ and $\mathbf{H}$ into their external (ext) and scattered (scat) field contributions, the electric component of the tensor defined in Eq. (2.24) is rewritten as

$$\overleftrightarrow{\mathcal{T}}^{\mathrm{E}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\left(\vec{\mathbf{E}}_{\mathrm{ext}} + \vec{\mathbf{E}}_{\mathrm{scat}}\right)\left(\vec{\mathbf{E}}^*_{\mathrm{ext}} + \vec{\mathbf{E}}^*_{\mathrm{scat}}\right) - \frac{\overleftrightarrow{\mathbf{I}}}{2}\left(\vec{\mathbf{E}}_{\mathrm{ext}} + \vec{\mathbf{E}}_{\mathrm{scat}}\right)\cdot\left(\vec{\mathbf{E}}^*_{\mathrm{ext}} + \vec{\mathbf{E}}^*_{\mathrm{scat}}\right)\right]$$

expanding the terms, we obtain

$$\overleftrightarrow{\mathfrak{T}}^{E} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\Bigg[\left(\vec{\mathbf{E}}_{\mathrm{scat}}\vec{\mathbf{E}}^{*}_{\mathrm{scat}} + \vec{\mathbf{E}}_{\mathrm{scat}}\vec{\mathbf{E}}^{*}_{\mathrm{ext}} + \vec{\mathbf{E}}_{\mathrm{ext}}\vec{\mathbf{E}}^{*}_{\mathrm{scat}} + \vec{\mathbf{E}}_{\mathrm{ext}}\vec{\mathbf{E}}^{*}_{\mathrm{ext}}\right) - \frac{\overleftrightarrow{\mathbf{I}}}{2}\left(\vec{\mathbf{E}}_{\mathrm{scat}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{scat}} + \vec{\mathbf{E}}_{\mathrm{scat}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{ext}} + \vec{\mathbf{E}}_{\mathrm{ext}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{scat}} + \vec{\mathbf{E}}_{\mathrm{ext}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{ext}}\right)\Bigg], \tag{2.39}$$

where the electric component of the stress tensor can be written as

$$\overleftrightarrow{\mathfrak{T}}^{E} = \overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{ss}} + \overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{int}} + \overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{ee}}, \tag{2.40}$$

with

$$\overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{ss}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\vec{\mathbf{E}}_{\mathrm{scat}}\vec{\mathbf{E}}^{*}_{\mathrm{scat}} - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}_{\mathrm{scat}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{scat}}\right], \tag{2.41}$$

$$\overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{ee}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\vec{\mathbf{E}}_{\mathrm{ext}}\vec{\mathbf{E}}^{*}_{\mathrm{ext}} - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}_{\mathrm{ext}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{ext}}\right], \tag{2.42}$$

$$\overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{int}} = \overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{se}} + \overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{es}}, \tag{2.43}$$

where

$$\overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{es}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\vec{\mathbf{E}}_{\mathrm{ext}}\vec{\mathbf{E}}^{*}_{\mathrm{scat}} - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}_{\mathrm{ext}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{scat}}\right], \tag{2.44}$$

$$\overleftrightarrow{\mathfrak{T}}^{E}_{\mathrm{se}} = \frac{\epsilon_0}{(4\pi\epsilon_0)}\mathrm{Re}\left[\vec{\mathbf{E}}_{\mathrm{scat}}\vec{\mathbf{E}}^{*}_{\mathrm{ext}} - \frac{\overleftrightarrow{\mathbf{I}}}{2}\vec{\mathbf{E}}_{\mathrm{scat}}\cdot\vec{\mathbf{E}}^{*}_{\mathrm{ext}}\right]. \tag{2.45}$$

Similarly, by substituting $\epsilon_0 \to \mu_0$ and $\vec{\mathbf{E}} \to \vec{\mathbf{H}}$ in Eqs. (2.39) to (2.45), the magnetic contributions to the tensor $\overleftrightarrow{\mathfrak{T}}$ are obtained, denoted as $\overleftrightarrow{\mathfrak{T}}^{H}_{\mathrm{ij}}$ where ij take the values {e, s}. The term $\overleftrightarrow{\mathfrak{T}}_{\mathrm{int}}$ can be interpreted as the component associated with the interaction of the electromagnetic field of the electron with the charges and currents induced in the nanoparticle (NP). If there were no NP, the electron's movement would remain unaffected, meaning it would not lose or transfer energy, linear momentum, or angular momentum ($\Delta L = 0$). In this scenario, the scattered electromagnetic fields would be zero, implying that the only contribution to the angular momentum would come from the component $\overleftrightarrow{\mathfrak{T}}_{\mathrm{ee}}$. Therefore, in general, it is concluded that the angular momentum transfer due to $\overleftrightarrow{\mathfrak{T}}_{\mathrm{ee}}$ is null [29]. It is also worth mentioning that the component $\overleftrightarrow{\mathfrak{T}}_{\mathrm{ss}}$, which depends solely on the electromagnetic fields scattered by the NP, is related to the interaction of the NP with itself, referred to as radiation reaction [32].

The next Section presents the analytical expressions for the external electromagnetic fields (produced by the electron) as well as those scattered by the NP.

2.2 Angular Momentum Transfer from a Fast Electron to a Spherical Nanoparticle

The external electromagnetic field produced by a fast electron, considered as a point charge $q = -e$, traveling at a constant velocity $\vec{\mathbf{v}} = v\hat{z}$ along the z-axis and following the trajectory $\vec{\mathbf{r}} = (b, 0, vt)$ (see Fig. 2.1), can be obtained via a Lorentz transformation from a reference frame in which the electron is at rest to a frame in which the electron is moving at a constant velocity $\vec{\mathbf{v}}$. This results in the following expressions for the fields [32]:

$$\vec{\mathbf{E}}_{\rm ext}(\vec{\mathbf{r}}, t) = (4\pi\epsilon_0)\frac{-e}{4\pi\epsilon_0}\frac{\gamma\left[\vec{\mathbf{R}} + (z - vt)\hat{z}\right]}{\left[R^2 + \gamma^2(z - vt)^2\right]^{3/2}}, \tag{2.46}$$

$$\vec{\mathbf{H}}_{\rm ext}(\vec{\mathbf{r}}, t) = (4\pi)\frac{-e}{4\pi}\frac{\gamma\,\vec{\mathbf{v}} \times \vec{\mathbf{R}}}{\left[R^2 + \gamma^2(z - vt)^2\right]^{3/2}}, \tag{2.47}$$

where c is the speed of light, $\gamma = \left(1 - \beta^2\right)^{-1/2}$, $\beta = v/c$, $\vec{\mathbf{R}} = (x - b)\hat{x} + y\hat{y}$, $R = \sqrt{(x - b)^2 + y^2}$, and $\vec{\mathbf{v}} \times \vec{\mathbf{R}} = v\left[(x - b)\hat{y} - y\hat{x}\right]$. The external electromagnetic fields can also be expressed in the frequency domain using a Fourier transform of Eqs. (2.46) and (2.47) [33]:

$$\boxed{\begin{aligned}\vec{\mathbf{E}}_{\rm ext}(\vec{\mathbf{r}}, \omega) &= (4\pi\epsilon_0)\frac{-e}{4\pi\epsilon_0}\frac{2\omega}{v^2\gamma}\mathrm{e}^{\mathrm{i}\omega(z/v)}\left\{\mathrm{sign}(\omega)K_1\left(\frac{|\omega|R}{v\gamma}\right)\hat{R} - \frac{\mathrm{i}}{\gamma}K_0\left(\frac{|\omega|R}{v\gamma}\right)\hat{z}\right\}, &(2.48)\\ \vec{\mathbf{H}}_{\rm ext}(\vec{\mathbf{r}}, \omega) &= (4\pi)\frac{-e}{4\pi}\frac{2e}{vc\gamma}|\omega|\,\mathrm{e}^{\mathrm{i}\omega z/v}K_1\left(\frac{|\omega|R}{v\gamma}\right)\hat{v} \times \hat{R}, &(2.49)\end{aligned}}$$

where K_ν is the modified Bessel function of the second kind of order ν and $\hat{v} = \vec{\mathbf{v}}/v$. Eqs. (2.48) and (2.48) are closed-form expressions with cylindrical symmetry. Since the calculation of the angular momentum transfer (AMT) requires the scattered fields by the nanoparticle (NP), which exhibit spherical symmetry, it is convenient to express the electromagnetic fields produced by the electron in a spherical basis.

The electric field produced by the electron can be derived using the time-dependent Green's function [33]

$$\vec{\mathbf{E}}_{\rm ext}(\vec{\mathbf{r}}, \omega) = e\left(\nabla - \mathrm{i}\frac{k\vec{\mathbf{v}}}{c}\right)\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}\omega t}G_0\left(\vec{\mathbf{r}} - \vec{\mathbf{r}}_t\right)\,dt, \tag{2.50}$$

where $G_0\left(\vec{\mathbf{r}} - \vec{\mathbf{r}}_t\right)$ is given by

$$G_0\left(\vec{\mathbf{r}} - \vec{\mathbf{r}}_t\right) = \frac{(4\pi\epsilon_0)}{4\pi\epsilon_0}\frac{\mathrm{e}^{\mathrm{i}k|\vec{\mathbf{r}} - \vec{\mathbf{r}}_t|}}{|\vec{\mathbf{r}} - \vec{\mathbf{r}}_t|}, \tag{2.51}$$

with $k = \omega/c$ being the wave number in vacuum, $\vec{\mathbf{r}}_t = \vec{\mathbf{r}}_0 + \vec{\mathbf{v}}t$ representing the position of the electron at time t, and $\vec{\mathbf{r}}_0 = (b, 0, 0)$. Expanding the Green's function in spherical coordinates yields [17]

$$G_0(\vec{\mathbf{r}} - \vec{\mathbf{r}}_t) = \frac{(4\pi\epsilon_0)}{\epsilon_0}k\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell}j_\ell(kr)h_\ell^+(kr_t)Y_{\ell,m}(\Omega_r)Y_{\ell,m}(\Omega_{r_t})^*, \tag{2.52}$$

where $Y_{\ell,m}$ are the scalar spherical harmonics, Ω_r is the solid angle of vector $\vec{\mathbf{r}}$, $h_\ell^+(x) = \mathrm{i}h_\ell^1(x)$ is the spherical Hankel function of the first kind of order ℓ, and $j_\ell(x)$ is the spherical Bessel function of order ℓ

[34]. By substituting Eq. (2.52) into Eq. (2.50), we obtain

$$\vec{\mathbf{E}}_{\text{ext}}\left(\vec{\mathbf{r}},t\right)=\left(\nabla-\mathrm{i}\frac{k\vec{\mathbf{v}}}{c}\right)\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell}j_\ell(kr)Y_{\ell,m}(\Omega_r)\phi_{\ell,m}, \tag{2.53}$$

where

$$\phi_{\ell,m}=\frac{(4\pi\epsilon_0)}{\epsilon_0}k\int_{-\infty}^{\infty}e^{i\omega t}h_\ell^{+}(kr_t)Y^{*}_{\ell,m}(\Omega_{r_t})\,dt. \tag{2.54}$$

The detailed analytical calculation of Eq. (2.54) can be found in Ref. [35]. Consequently, the external electromagnetic field in spherical multipolar representation can be expressed as:

Multipolar expansion of the external electromagnetic field produced by the swift electron

$$\vec{\mathbf{E}}_{\text{ext}}=\sum_{\ell,m}\left(\mathscr{E}^{\text{er}}_{\ell,m}\hat{r}+\mathscr{E}^{\text{e}\theta}_{\ell,m}\hat{\theta}+\mathscr{E}^{\text{e}\varphi}_{\ell,m}\hat{\varphi}\right), \tag{2.55}$$

$$\vec{\mathbf{H}}_{\text{ext}}=\sum_{\ell,m}\left(\mathscr{H}^{\text{er}}_{\ell,m}\hat{r}+\mathscr{H}^{\text{e}\theta}_{\ell,m}\hat{\theta}+\mathscr{H}^{\text{e}\varphi}_{\ell,m}\hat{\varphi}\right), \tag{2.56}$$

where

$$\mathscr{E}^{\text{er}}_{\ell,m}=\mathrm{e}^{\mathrm{i}m\varphi}D^{\text{ext}}_{\ell,m}\ell(\ell+1)P_\ell^m\left(\cos\theta\right)\frac{j_\ell\left(kr\right)}{kr}, \tag{2.57}$$

$$\begin{aligned}\mathscr{E}^{\text{e}\theta}_{\ell,m}=&-\mathrm{e}^{\mathrm{i}m\varphi}C^{\text{ext}}_{\ell,m}\frac{m}{\sin\theta}j_\ell\left(kr\right)P_\ell^m(\cos\theta)-\mathrm{e}^{\mathrm{i}m\varphi}D^{\text{ext}}_{\ell,m}\left[(\ell+1)\frac{\cos\theta}{\sin\theta}P_\ell^m\left(\cos\theta\right)-\right.\\&\left.\frac{(\ell-m+1)}{\sin\theta}P^m_{\ell+1}\left(\cos\theta\right)\right]\left[(\ell+1)\frac{j_\ell\left(kr\right)}{kr}-j_{\ell+1}\left(kr\right)\right],\end{aligned} \tag{2.58}$$

$$\begin{aligned}\mathscr{E}^{\text{e}\varphi}_{\ell,m}=&\,\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi}C^{\text{ext}}_{\ell,m}j_\ell\left(kr\right)\left[(\ell+1)\frac{\cos\theta}{\sin\theta}P_\ell^m\left(\cos\theta\right)-\frac{\ell-m+1}{\sin\theta}P^m_{\ell+1}\left(\cos\theta\right)\right]\\&+\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi}D^{\text{ext}}_{\ell,m}\frac{m}{\sin\theta}P_\ell\left(\cos\theta\right)\left[(\ell+1)\frac{j_\ell\left(kr\right)}{kr}-j_{\ell+1}(kr)\right],\end{aligned} \tag{2.59}$$

and

$$\mathscr{H}^{\text{er}}_{\ell,m}=\mathrm{e}^{\mathrm{i}m\varphi}C^{\text{ext}}_{\ell,m}\ell(\ell+1)P_\ell^m\left(\cos\theta\right)\frac{j_\ell\left(kr\right)}{kr}, \tag{2.60}$$

$$\begin{aligned}\mathscr{H}^{\text{e}\theta}_{\ell,m}=&\,\mathrm{e}^{\mathrm{i}m\varphi}D^{\text{ext}}_{\ell,m}\frac{m}{\sin\theta}j_\ell\left(kr\right)P_\ell^m(\cos\theta)-\mathrm{e}^{\mathrm{i}m\varphi}C^{\text{ext}}_{\ell,m}\left[(\ell+1)\frac{\cos\theta}{\sin\theta}P_\ell^m\left(\cos\theta\right)-\right.\\&\left.\frac{(\ell-m+1)}{\sin\theta}P^m_{\ell+1}\left(\cos\theta\right)\right]\left[(\ell+1)\frac{j_\ell\left(kr\right)}{kr}-j_{\ell+1}\left(kr\right)\right],\end{aligned} \tag{2.61}$$

$$\begin{aligned}\mathscr{H}^{\text{e}\varphi}_{\ell,m}=&-\,\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi}D^{\text{ext}}_{\ell,m}j_\ell\left(kr\right)\left[(\ell+1)\frac{\cos\theta}{\sin\theta}P_\ell^m\left(\cos\theta\right)-\frac{\ell-m+1}{\sin\theta}P^m_{\ell+1}\left(\cos\theta\right)\right]\\&+\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi}C^{\text{ext}}_{\ell,m}\frac{m}{\sin\theta}P_\ell\left(\cos\theta\right)\left[(\ell+1)\frac{j_\ell\left(kr\right)}{kr}-j_{\ell+1}(kr)\right],\end{aligned} \tag{2.62}$$

where P_ℓ^m are the associated Legendre functions, and the scalar coefficients $C^{\rm ext}_{\ell,m}$ and $D^{\rm ext}_{\ell,m}$ are given by:

$$\boxed{\begin{aligned} C^{\rm ext}_{\ell,m} &= {\rm i}^\ell \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\psi^{\rm M,ext}_{\ell,m}, \qquad (2.63)\\ D^{\rm ext}_{\ell,m} &= {\rm i}^\ell \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\psi^{\rm E,ext}_{\ell,m}, \qquad (2.64)\end{aligned}}$$

with $\psi^{E,\rm ext}_{\ell,m}$ and $\psi^{M,\rm ext}_{\ell,m}$ being the coefficients in the spherical representation of the auxiliary potentials, defined in Ref. [35].

2.3 Electromagnetic Fields Scattered by the Nanoparticle

The electromagnetic fields satisfy the source-free Helmholtz equation:

$$\left(\nabla^2 + k^2\right)\vec{\mathbf{E}} = \vec{\mathbf{0}}, \tag{2.65}$$

$$\left(\nabla^2 + k^2\right)\vec{\mathbf{H}} = \vec{\mathbf{0}}. \tag{2.66}$$

The solutions to Eqs. (2.65) and (2.66) can be written as [36]:

$$\vec{\mathbf{E}} = \nabla\psi^{\rm L} + \vec{\mathbf{L}}\psi^{\rm M} - \frac{\rm i}{k}\nabla\times\vec{\mathbf{L}}\psi^{\rm E}, \tag{2.67}$$

$$\vec{\mathbf{H}} = -\frac{\rm i}{k}\nabla\times\vec{\mathbf{L}}\psi^{M} - \vec{\mathbf{L}}\psi^{\rm E}, \tag{2.68}$$

where $\vec{\mathbf{L}} = -{\rm i}\vec{\mathbf{r}}\times\nabla$ is the orbital angular momentum operator, and $\psi^{\rm L}$, $\psi^{\rm E}$, and $\psi^{\rm M}$ are scalar functions that satisfy the scalar Helmholtz equation. Since the external electric field is solenoidal ($\nabla\cdot\vec{\mathbf{E}} = 0$), the scalar function $\psi^{\rm L}$ must be zero. The scalar functions $\psi^{\rm E}$ and $\psi^{\rm M}$, for both the external and scattered fields, can be written in the spherical basis defined by the coordinate system shown in Fig. 2.1, as [17]:

$$\psi^{\rm E,ext}(\vec{\mathbf{r}}) = \sum_{\ell=1}^{\infty}\sum_{m=-\ell}^{\ell} {\rm i}^\ell j_\ell\left(kr\right) Y_{\ell,m}(\Omega_r)\psi^{\rm E,ext}_{\ell,m}, \tag{2.69}$$

$$\psi^{\rm M,ext}(\vec{\mathbf{r}}) = \sum_{\ell=1}^{\infty}\sum_{m=-\ell}^{\ell} {\rm i}^\ell j_\ell\left(kr\right) Y_{\ell,m}(\Omega_r)\psi^{\rm M,ext}_{\ell,m}, \tag{2.70}$$

$$\psi^{\rm E,scat}(\vec{\mathbf{r}}) = \sum_{\ell=1}^{\infty}\sum_{m=-\ell}^{\ell} {\rm i}^\ell h^+_\ell\left(kr\right) Y_{\ell,m}(\Omega_r)\psi^{\rm E,scat}_{\ell,m}, \tag{2.71}$$

$$\psi^{\rm M,scat}(\vec{\mathbf{r}}) = \sum_{\ell=1}^{\infty}\sum_{m=-\ell}^{\ell} {\rm i}^\ell h^+_\ell\left(kr\right) Y_{\ell,m}(\Omega_r)\psi^{\rm M,scat}_{\ell,m}. \tag{2.72}$$

Applying the boundary conditions for the electromagnetic fields on a spherical particle, the scattered scalar electromagnetic potentials by the nanoparticle (NP) can be expressed in terms of the external scalar electromagnetic potentials as follows [17]:

$$\psi_{\ell,m}^{\mathrm{E,scat}} = t_\ell^{\mathrm{E}} \psi_{\ell,m}^{\mathrm{E,ext}}, \tag{2.73}$$

$$\psi_{\ell,m}^{\mathrm{M,scat}} = t_\ell^{\mathrm{M}} \psi_{\ell,m}^{\mathrm{M,ext}}, \tag{2.74}$$

where the coefficients t_ℓ^{E} and t_ℓ^{M} for homogeneous spherical particles correspond to the Mie coefficients [37]:

$$t_\ell^{\mathrm{E}} = \frac{-j_\ell(x_0)\,[x_i j_\ell(x_i)]' + \epsilon\, j_\ell(x_i)\,[x_0 j_\ell(x_0)]'}{h_\ell^+(x_0)\,[x_i j_l(x_i)]' - \epsilon j_l(x_i)\,\left[x_0 h_\ell^+(x_0)\right]'}, \tag{2.75}$$

$$t_\ell^{\mathrm{M}} = \frac{-x_i j_\ell(x_0)\, j_\ell'(x_i) + x_0 j_\ell'(x_0)\, j_\ell(x_i)}{x_i h_\ell^+(x_0)\, j_\ell'(x_i) - x_0 h_\ell^{+\prime}(x_0)\, j_\ell(x_i)}, \tag{2.76}$$

where $x_0 = ka$ and $x_i = ka\sqrt{\epsilon}$, with a being the particle radius and ϵ the dielectric function of the NP. The primes in Eqs. (2.75) and (2.76) denote derivatives with respect to the argument.

By substituting the scattered scalar potentials from Eqs. (2.73) and (2.74) into Eqs. (2.67) and (2.68), we obtain:

Multipolar expansion of the electromagnetic field scattered by the NP

$$\vec{\mathbf{E}}_{\mathrm{scat}} = \sum_{\ell,m} \left(\mathscr{E}_{\ell,m}^{\mathrm{sr}} \hat{r} + \mathscr{E}_{\ell,m}^{\mathrm{s}\theta} \hat{\theta} + \mathscr{E}_{\ell,m}^{\mathrm{s}\varphi} \hat{\varphi} \right), \tag{2.77}$$

$$\vec{\mathbf{H}}_{\mathrm{scat}} = \sum_{\ell,m} \left(\mathscr{H}_{\ell,m}^{\mathrm{sr}} \hat{r} + \mathscr{H}_{\ell,m}^{\mathrm{s}\theta} \hat{\theta} + \mathscr{H}_{\ell,m}^{\mathrm{s}\varphi} \hat{\varphi} \right), \tag{2.78}$$

where

$$\mathscr{E}_{\ell,m}^{\mathrm{sr}} = \mathrm{e}^{\mathrm{i}m\varphi} D_{\ell,m}^{\mathrm{scat}} \ell(\ell+1) P_\ell^m(\cos\theta)\, \frac{h_\ell^+(k_0 r)}{k_0 r}, \tag{2.79}$$

$$\begin{aligned}\mathscr{E}_{\ell,m}^{\mathrm{s}\theta} =& -\mathrm{e}^{\mathrm{i}m\varphi} C_{\ell,m}^{\mathrm{scat}} \frac{m}{\sin\theta} h_\ell^+(k_0 r)\, P_\ell^m(\cos\theta) - \mathrm{e}^{\mathrm{i}m\varphi} D_{\ell,m}^{\mathrm{scat}} \Bigg[(\ell+1)\, \frac{\cos\theta}{\sin\theta} P_\ell^m(\cos\theta) - \\ & \frac{(\ell-m+1)}{\sin\theta} P_{\ell+1}^m(\cos\theta) \Bigg] \left[(\ell+1)\, \frac{h_\ell^+(k_0 r)}{k_0 r} - h_{\ell+1}^+(k_0 r) \right],\end{aligned} \tag{2.80}$$

$$\begin{aligned}\mathscr{E}_{\ell,m}^{\mathrm{s}\varphi} =& \,\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi} C_{\ell,m}^{\mathrm{scat}}\, h_\ell^+(k_0 r) \left[(\ell+1)\, \frac{\cos\theta}{\sin\theta} P_\ell^m(\cos\theta) - \frac{\ell-m+1}{\sin\theta} P_{\ell+1}^m(\cos\theta) \right] \\ & + \mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi} D_{\ell,m}^{\mathrm{scat}} \frac{m}{\sin\theta} P_\ell(\cos\theta) \left[(\ell+1)\, \frac{h_\ell^+(k_0 r)}{k_0 r} - h_{\ell+1}^+(k_0 r) \right],\end{aligned} \tag{2.81}$$

and

$$\mathscr{H}^{\mathrm{sr}}_{\ell,m} = \mathrm{e}^{\mathrm{i}m\varphi} C^{\mathrm{scat}}_{\ell,m} \ell(\ell+1) P^{m}_{\ell}(\cos\theta) \frac{h^{+}_{\ell}(k_0 r)}{k_0 r}, \tag{2.82}$$

$$\begin{aligned}\mathscr{H}^{\mathrm{s}\theta}_{\ell,m} =& \mathrm{e}^{\mathrm{i}m\varphi} D^{\mathrm{scat}}_{\ell,m} \frac{m}{\sin\theta} h^{+}_{\ell}(k_0 r) P^{m}_{\ell}(\cos\theta) - \mathrm{e}^{\mathrm{i}m\varphi} C^{\mathrm{scat}}_{\ell,m} \left[(\ell+1) \frac{\cos\theta}{\sin\theta} P^{m}_{\ell}(\cos\theta) - \right. \\ & \left. \frac{(\ell-m+1)}{\sin\theta} P^{m}_{\ell+1}(\cos\theta) \right] \left[(\ell+1) \frac{h^{+}_{\ell}(k_0 r)}{k_0 r} - h^{+}_{\ell+1}(k_0 r) \right],\end{aligned} \tag{2.83}$$

$$\begin{aligned}\mathscr{H}^{\mathrm{s}\varphi}_{\ell,m} =& -\mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi} D^{\mathrm{scat}}_{\ell,m} h^{+}_{\ell}(k_0 r) \left[(\ell+1) \frac{\cos\theta}{\sin\theta} P^{m}_{\ell}(\cos\theta) - \frac{\ell-m+1}{\sin\theta} P^{m}_{\ell+1}(\cos\theta) \right] \\ & + \mathrm{i}\,\mathrm{e}^{\mathrm{i}m\varphi} C^{\mathrm{scat}}_{\ell,m} \frac{m}{\sin\theta} P_{\ell}(\cos\theta) \left[(\ell+1) \frac{h^{+}_{\ell}(k_0 r)}{k_0 r} - h^{+}_{\ell+1}(k_0 r) \right],\end{aligned} \tag{2.84}$$

with the scalar coefficients $C^{\mathrm{scat}}_{\ell,m}$ and $D^{\mathrm{scat}}_{\ell,m}$ given by

$$C^{\mathrm{scat}}_{\ell,m} = \mathrm{i}^{\ell} \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}}\, t^{\mathrm{M}}_{\ell} \psi^{\mathrm{M,ext}}_{\ell,m}, \tag{2.85}$$

$$D^{\mathrm{scat}}_{\ell,m} = \mathrm{i}^{\ell} \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}}\, t^{\mathrm{E}}_{\ell} \psi^{\mathrm{E,ext}}_{\ell,m}. \tag{2.86}$$

Once the total electromagnetic field has been computed, one can determine the tensor $\overleftrightarrow{\mathfrak{T}}$ as given by Eq. (2.24), and consequently, compute the spectral density of TMA, expressed by Eq. (2.25):

$$\mathcal{L}_i(\omega) = \frac{1}{\pi} \oint_S \epsilon_i{}^{lj} r_l \mathfrak{T}_{jk}(\vec{\mathbf{r}}, \omega)\, n^k \, dS.$$

3

Progress in the Analytical Solution for the Spectral Density of Angular Momentum Transfer to Spherical Nanoparticles

This Section presents the results of the semi-analytical expressions that describe the spectral contribution of angular momentum transfer (AMT) in the interaction between a fast electron and a spherical nanoparticle (NP). Specifically, the spectral densities of the AMT are developed through the Fourier transform of the Maxwell stress tensor, expanding the cross terms of the electromagnetic fields, and calculating the integrals semi-analytically over a spherical surface S enclosing the NP.

It is important to note that to obtain the AMT from the semi-analytical expressions developed for the spectral contributions to the AMT, a numerical integration over the entire frequency space is required. For this purpose, the Gauss-Kronrod quadrature method will be used, which allows for the estimation of the numerical error incurred [38].

3.1 Spectral Density of Angular Momentum Transfer

In Section 2, an expression for the AMT from a fast electron to a spherical NP was obtained, given by Eq. (2.26),

$$\Delta\vec{\mathbf{L}} = \int_0^\infty \vec{\mathcal{L}}\left(\omega\right)\, d\omega.$$

where the spectral density of angular momentum is defined in Eq. (2.25)

$$\mathcal{L}_i\left(\omega\right) = \frac{1}{\pi}\oint_S \epsilon_i^{\ lj} r_l \mathcal{T}_{jk}\left(\vec{\mathbf{r}},\omega\right) n^k \, dS,$$

so that the components of the AMT are given by

$$\mathcal{L}_y = \frac{R^3}{\pi}\int_0^\pi\int_0^{2\pi} \sin\theta\left[\cos\varphi\,\mathcal{T}_{\theta r} - \cos\theta\sin\varphi\,\mathcal{T}_{\varphi r}\right]\, d\varphi\, d\theta,\text{[1]} \tag{3.1}$$

[1]The term y is highlighted in cyan because it is the only non-vanishing term, as will be demonstrated later. This color coding will be consistently applied to emphasize non-vanishing terms for the reader's convenience.

$$\mathcal{L}_x = \frac{R^3}{\pi} \int_0^{\pi} \int_0^{2\pi} \sin\theta \left[\sin\varphi\, \mathcal{T}_{\theta r} + \cos\theta \cos\varphi\, \mathcal{T}_{\varphi r}\right] d\varphi\, d\theta, \tag{3.2}$$

$$\mathcal{L}_z = \frac{R^3}{\pi} \int_0^{\pi} \int_0^{2\pi} \sin^2\theta\, \mathcal{T}_{\varphi r}\, d\varphi\, d\theta, \tag{3.3}$$

where

$$\mathcal{T}_{\theta r} = \mathrm{Re}\left[\frac{\epsilon_0}{(4\pi\epsilon_0)} E_\theta(\vec{\mathbf{r}},\omega)\, E_r^*(\vec{\mathbf{r}},\omega) + \frac{\mu_0}{(4\pi\mu_0)} H_\theta(\vec{\mathbf{r}},\omega)\, H_r^*(\vec{\mathbf{r}},\omega)\right], \tag{3.4}$$

$$\mathcal{T}_{\varphi r} = \mathrm{Re}\left[\frac{\epsilon_0}{(4\pi\epsilon_0)} E_\varphi(\vec{\mathbf{r}},\omega)\, E_r^*(\vec{\mathbf{r}},\omega) + \frac{\mu_0}{(4\pi\mu_0)} H_\varphi(\vec{\mathbf{r}},\omega)\, H_r^*(\vec{\mathbf{r}},\omega)\right]. \tag{3.5}$$

It is important to note that $E_i = E_i^{\mathrm{ext}} + E_i^{\mathrm{scat}}$ and $H_i = H_i^{\mathrm{ext}} + H_i^{\mathrm{scat}}$ can be separated, and the terms $E_\alpha(\omega)E_r^*(\omega)$ and $H_\alpha(\omega)H_r^*(\omega)$ can be expanded in terms of the multipolar sums expressed in Eqs. (2.55), (2.56), (2.77), and (2.78), as follows:

$$E_\alpha^{\mathrm{a}}(\vec{\mathbf{r}},\omega)\, E_r^{\mathrm{b}\,*}(\vec{\mathbf{r}},\omega) = \sum_{\ell,m}\sum_{\ell',m'} \mathscr{E}^{\mathrm{a}\alpha}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'}, \tag{3.6}$$

$$H_\alpha^{\mathrm{a}}(\vec{\mathbf{r}},\omega)\, H_r^{\mathrm{b}\,*}(\vec{\mathbf{r}},\omega) = \sum_{\ell,m}\sum_{\ell',m'} \mathscr{H}^{\mathrm{a}\alpha}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}, \tag{3.7}$$

where a and b can take the values “scat” (denoting scattered field components) or “ext” (external field components), and α corresponds to the projection in $\hat{\theta}$ or $\hat{\varphi}$.

The term $\mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'}$ is given by

$$\begin{aligned}\mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{ &- C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_\ell Z^{\mathrm{b}\,*}_{\ell'}}{kr} m \frac{P^m_\ell P^{m'}_{\ell'}}{\sin\theta} - D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_\ell Z^{\mathrm{b}\,*}_{\ell'}}{kr} \\ &\times \left[(\ell+1)\frac{P^m_\ell P^{m'}_{\ell'}\cos\theta}{\sin\theta} - (l-m+1)\frac{P^m_{\ell+1}P^{m'}_{\ell'}}{\sin\theta}\right]\Bigg\} \ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi},\end{aligned} \tag{3.8}$$

where $k = \omega/c$, P^m_ℓ is the associated Legendre function of degree ℓ and order m, the coefficients $C^a_{\ell,m}$ and $D^a_{\ell,m}$ are defined in Eqs. (2.63), (2.64), (2.85) and (2.86), $Z^{\mathrm{s}}_\ell = h^+_\ell(kr)$ for the scattered field, and $Z^{\mathrm{e}}_\ell = j_\ell(kr)$ for the external field, and we have defined:

$$f^{\mathrm{a}}_\ell = (\ell+1)\frac{Z^{\mathrm{a}}_\ell}{kr} - Z^{\mathrm{a}}_{\ell+1}. \tag{3.9}$$

Similarly, the term $\mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}$ can be written as

$$\begin{aligned}\mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{ &D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_\ell Z^{\mathrm{b}\,*}_{\ell'}}{kr} m \frac{P^m_\ell P^{m'}_{\ell'}}{\sin\theta} - C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_\ell Z^{\mathrm{b}\,*}_{\ell'}}{kr} \\ &\times \left[(\ell+1)\frac{P^m_\ell P^{m'}_{\ell'}\cos\theta}{\sin\theta} - (l-m+1)\frac{P^m_{\ell+1}P^{m'}_{\ell'}}{\sin\theta}\right]\Bigg\} \ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}.\end{aligned} \tag{3.10}$$

Analogously, the term $\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'}$ can be written as

$$\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{C^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\left[(\ell+1)\,\frac{P^m_{\ell}P^{m'}_{\ell'}\cos\theta}{\sin\theta} - (\ell-m+1)\,\frac{P^m_{\ell+1}P_{\ell'}}{\sin\theta}\right] + D^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\,\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sin\theta}\Bigg\}\,\mathrm{i}\,\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}, \tag{3.11}$$

and finally, the term $\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}$ can be written as

$$\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{-D^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\left[(\ell+1)\,\frac{P^m_{\ell}P^{m'}_{\ell'}\cos\theta}{\sin\theta} - (\ell-m+1)\,\frac{P^m_{\ell+1}P_{\ell'}}{\sin\theta}\right] + C^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\,\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sin\theta}\Bigg\}\,\mathrm{i}\,\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}. \tag{3.12}$$

To simplify the alegraic development, the variable change $x=\cos\theta$ is propoused, an the following recurrence relation is taken into account

$$(1-x^2)\frac{dP^m_{\ell}}{dx} = (\ell+1)xP^m_{\ell} - (l-m+1)P^m_{\ell+1}, \tag{3.13}$$

and thus we can rewrite the following terms as

$$\mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{-C^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}m\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sqrt{1-x^2}} - D^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr} \times\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_{\ell}}{dx}\Bigg\}\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}, \tag{3.14}$$

$$\mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{D^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}m\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sqrt{1-x^2}} - C^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr} \times\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_{\ell}}{dx}\Bigg\}\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}, \tag{3.15}$$

$$\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{C^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_{\ell}}{dx} + D^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\,\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sqrt{1-x^2}}\Bigg\}\,\mathrm{i}\,\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}, \tag{3.16}$$

$$\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} = \Bigg\{-D^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_{\ell}}{dx} + C^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kr}\,\frac{P^m_{\ell}P^{m'}_{\ell'}}{\sqrt{1-x^2}}\Bigg\}\,\mathrm{i}\,\ell'(\ell'+1)\mathrm{e}^{\mathrm{i}(m-m')\varphi}. \tag{3.17}$$

The integrals over φ can be calculated considering that

$$\int_0^{2\pi} \mathrm{e}^{\mathrm{i}(m-m')\varphi} \cos\varphi \, d\varphi = \pi \left(\delta_{m+1,m'} + \delta_{m-1,m'}\right), \tag{3.18}$$

$$\int_0^{2\pi} \mathrm{e}^{\mathrm{i}(m-m')\varphi} \sin\varphi \, d\varphi = \mathrm{i}\pi \left(\delta_{m-1,m'} - \delta_{m+1,m'}\right), \tag{3.19}$$

where $\delta_{i,j}$ is the Kronecker delta.

To obtain the angular momentum transfer (AMT), it is necessary to integrate the spectral density $\mathcal{L}$ over the frequency space. To calculate the spectral density, the surface integrals over the spherical shell S enclosing the NP must be solved [see Eqs. (3.2)-(3.3)]. Therefore, it is required to compute the surface integral of the components $\mathfrak{T}_{\theta r}$ and $\mathfrak{T}_{\varphi r}$, described in Eqs. (3.4) and (3.5). By substituting Eqs. (3.6) and (3.7) into Eqs. (3.4) and (3.5), we obtain

$$\mathfrak{T}_{\theta r} = \sum_{a,b} \sum_{\ell,m} \sum_{\ell',m'} \mathrm{Re}\left[\frac{\epsilon_0}{(4\pi\epsilon_0)} \mathscr{E}^{\mathrm{a}\theta}_{\ell,m} \mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)} \mathscr{H}^{\mathrm{a}\theta}_{\ell,m} \mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\right], \tag{3.20}$$

$$\mathfrak{T}_{\varphi r} = \sum_{a,b} \sum_{\ell,m} \sum_{\ell',m'} \mathrm{Re}\left[\frac{\epsilon_0}{(4\pi\epsilon_0)} \mathscr{E}^{\mathrm{a}\varphi}_{\ell,m} \mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)} \mathscr{H}^{\mathrm{a}\varphi}_{\ell,m} \mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\right], \tag{3.21}$$

where the sum over a, b considers the interactions $\mathfrak{T}_{\mathrm{ss}}$, $\mathfrak{T}_{\mathrm{ee}}$, $\mathfrak{T}_{\mathrm{es}}$, and $\mathfrak{T}_{\mathrm{se}}$, described in Eqs. (2.41)-(2.45).

The integrals of the cross terms in the multipolar contribution to the electromagnetic fields' stress tensor $\overleftrightarrow{\mathfrak{T}}$, as shown in Eqs. (3.20) and (3.21), can be written as

$$\int_0^{4\pi} \mathscr{E}^{\mathrm{a}\theta}_{\ell,m} \mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} \sin\varphi \, d\Omega = \mathrm{i}\pi \left(\delta_{m-1,m'} - \delta_{m+1,m'}\right) \Bigg\{ - C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} m \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{dx}{\sqrt{1-x^2}} - D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} \sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx} \, dx \Bigg\} \ell'(\ell'+1), \tag{3.22}$$

$$\int_0^{4\pi} \mathscr{H}^{\mathrm{a}\theta}_{\ell,m} \mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \sin\varphi \, d\Omega = \mathrm{i}\pi \left(\delta_{m-1,m'} - \delta_{m+1,m'}\right) \Bigg\{ D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} m \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{dx}{\sqrt{1-x^2}} - C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} \sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx} \, dx \Bigg\} \ell'(\ell'+1), \tag{3.23}$$

$$\int_0^{4\pi} \mathscr{E}^{\mathrm{a}\theta}_{\ell,m} \mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} \cos\varphi \, d\Omega = \pi \left(\delta_{m-1,m'} + \delta_{m+1,m'}\right) \Bigg\{ - C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} m \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{dx}{\sqrt{1-x^2}} - D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} \sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx} \, dx \Bigg\} \ell'(\ell'+1), \tag{3.24}$$

$$\int_0^{4\pi} \mathscr{H}^{\mathrm{a}\theta}_{\ell,m} \mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \cos\varphi \, d\Omega = \pi \left(\delta_{m-1,m'} + \delta_{m+1,m'}\right) \Bigg\{ D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} m \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{dx}{\sqrt{1-x^2}} - C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} \sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx} \, dx \Bigg\} \ell'(\ell'+1), \tag{3.25}$$

$$\int_0^{4\pi} \mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} \cos\theta \sin\varphi \, d\Omega = -\,\pi\left(\delta_{m-1,m'} - \delta_{m+1,m'}\right)\Bigg\{ C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} x\sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{x}{\sqrt{1-x^2}}\, dx \Bigg\} \ell'(\ell'+1), \quad (3.26)$$

$$\int_0^{4\pi} \mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \cos\theta \sin\varphi \, d\Omega = -\,\pi\left(\delta_{m-1,m'} - \delta_{m+1,m'}\right)\Bigg\{ -D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} x\sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{x}{\sqrt{1-x^2}}\, dx \Bigg\} \ell'(\ell'+1), \quad (3.27)$$

$$\int_0^{4\pi} \mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} \cos\theta \cos\varphi \, d\Omega = \mathrm{i}\,\pi\left(\delta_{m-1,m'} + \delta_{m+1,m'}\right)\Bigg\{ C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} x\sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{x}{\sqrt{1-x^2}}\, dx \Bigg\} \ell'(\ell'+1), \quad (3.28)$$

$$\int_0^{4\pi} \mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \cos\theta \cos\varphi \, d\Omega = \mathrm{i}\,\pi\left(\delta_{m-1,m'} + \delta_{m+1,m'}\right)\Bigg\{ -D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} x\sqrt{1-x^2} P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} P^m_\ell(x) P^{m'}_{\ell'}(x) \frac{x}{\sqrt{1-x^2}}\, dx \Bigg\} \ell'(\ell'+1), \quad (3.29)$$

$$\int_0^{4\pi} \mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} \sin\theta \, d\Omega = \mathrm{i}\,2\pi\,\delta_{mm'}\Bigg\{ C^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} (1-x^2) P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + D^{\mathrm{a}}_{\ell,m} D^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \Delta_{\ell\ell'} \Bigg\} \ell'(\ell'+1), \quad (3.30)$$

$$\int_0^{4\pi} \mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \sin\theta \, d\Omega = \mathrm{i}\,2\pi\,\delta_{mm'}\Bigg\{ -D^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} \frac{Z^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \int_{-1}^{1} (1-x^2) P^{m'}_{\ell'} \frac{dP^m_\ell}{dx}\, dx + C^{\mathrm{a}}_{\ell,m} C^{\mathrm{b}\,*}_{\ell',m'} m \frac{f^{\mathrm{a}}_{\ell} Z^{\mathrm{b}\,*}_{\ell'}}{kR} \Delta_{\ell\ell'} \Bigg\} \ell'(\ell'+1). \quad (3.31)$$

For the particular case where only two associated Legendre functions are integrated, the orthogonality relation is used [34]

$$\int_{-1}^{1} P^m_\ell(x) P^m_{\ell'}(x)\, dx = \Delta_{\ell\ell'} \qquad \text{where} \qquad \Delta_{\ell\ell'} = \frac{2(\ell+m)!}{(2\ell+1)(\ell-m)!}\delta_{\ell\ell'}, \quad (3.32)$$

and defining the following quantities

$$IS^{\mathrm{a}\theta\mathrm{br}}_{\ell m\ell' m'} = \frac{R^3}{\pi}\int_0^{4\pi} \mathrm{Re}\left(\frac{\epsilon_0}{(4\pi\epsilon_0)} \mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)} \mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \right) \sin\varphi \, d\Omega = 0, \quad (3.33)$$

$$IC^{\mathrm{a}\theta\mathrm{br}}_{\ell m\ell' m'} = \frac{R^3}{\pi}\int_0^{4\pi} \mathrm{Re}\left(\frac{\epsilon_0}{(4\pi\epsilon_0)} \mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)} \mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \right) \cos\varphi \, d\Omega, \quad (3.34)$$

$$ICS^{\mathrm{a}\varphi\mathrm{br}}_{\ell m\ell' m'} = \frac{R^3}{\pi}\int_0^{4\pi} \mathrm{Re}\left(\frac{\epsilon_0}{(4\pi\epsilon_0)} \mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)} \mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'} \right) \cos\theta\sin\varphi \, d\Omega, \quad (3.35)$$

$$ICC^{\mathrm{a}\varphi\mathrm{br}}_{\ell m\ell' m'} = \frac{R^3}{\pi}\int_0^{4\pi}\mathrm{Re}\left(\frac{\epsilon_0}{(4\pi\epsilon_0)}\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)}\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\right)\cos\theta\cos\varphi\, d\Omega = 0, \tag{3.36}$$

$$IS^{\mathrm{a}\varphi\mathrm{br}}_{\ell m\ell' m'} = \frac{R^3}{\pi}\int_0^{4\pi}\mathrm{Re}\left(\frac{\epsilon_0}{(4\pi\epsilon_0)}\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'} + \frac{\mu_0}{(4\pi\mu_0)}\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\right)\sin\theta\, d\Omega = 0, \tag{3.37}$$

the Eqs. (3.1), (3.2) and (3.3) can be rewritten as

$$\mathcal{L}_x = 0, \tag{3.38}$$

$$\mathcal{L}_y = \sum_{\ell,m}\sum_{\ell',m'}\left(IC^{\mathrm{a}\theta\mathrm{br}}_{\ell m\ell' m'} - ICS^{\mathrm{a}\varphi\mathrm{br}}_{\ell m\ell' m'}\right), \tag{3.39}$$

$$\mathcal{L}_z = 0, \tag{3.40}$$

by taking the real part of the functions of interest and recognizing the purely imaginary contributions of the x and z components, we find that only the y component contributes to the spectral density of the angular momentum transfer. This result aligns with the previously stated fact that, due to the symmetry of the problem, the y component is the only non-zero component of the angular momentum transfer.

Therefore, the following discussion will only develop the terms related to the y component of the angular momentum transfer which are

$$\int_0^{4\pi}\mathscr{E}^{\mathrm{a}\theta}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'}\cos\varphi\, d\Omega = \pi\left(\delta_{m-1,m'} + \delta_{m+1,m'}\right)\Bigg\{ - C^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}m\int_{-1}^{1}P^{m}_{\ell}(x)P^{m'}_{\ell'}(x)\,\frac{dx}{\sqrt{1-x^2}}$$
$$- D^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_\ell}{dx}\,dx\Bigg\}\ell'(\ell'+1), \tag{3.41}$$

$$\int_0^{4\pi}\mathscr{H}^{\mathrm{a}\theta}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\cos\varphi\, d\Omega = \pi\left(\delta_{m-1,m'} + \delta_{m+1,m'}\right)\Bigg\{D^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}m\int_{-1}^{1}P^{m}_{\ell}(x)P^{m'}_{\ell'}(x)\,\frac{dx}{\sqrt{1-x^2}}$$
$$- C^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_\ell}{dx}\,dx\Bigg\}\ell'(\ell'+1), \tag{3.42}$$

$$\int_0^{4\pi}\mathscr{E}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{E}^{\mathrm{br}\,*}_{\ell',m'}\cos\theta\sin\varphi\, d\Omega = -\pi\left(\delta_{m-1,m'} - \delta_{m+1,m'}\right)\Bigg\{C^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}x\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_\ell}{dx}\,dx$$
$$+ D^{\mathrm{a}}_{\ell,m}D^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}P^{m}_{\ell}(x)P^{m'}_{\ell'}(x)\,\frac{x}{\sqrt{1-x^2}}\,dx\Bigg\}\ell'(\ell'+1), \tag{3.43}$$

$$\int_0^{4\pi}\mathscr{H}^{\mathrm{a}\varphi}_{\ell,m}\mathscr{H}^{\mathrm{br}\,*}_{\ell',m'}\cos\theta\sin\varphi\, d\Omega = -\pi\left(\delta_{m-1,m'} - \delta_{m+1,m'}\right)\Bigg\{ - D^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}\frac{Z^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}x\sqrt{1-x^2}P^{m'}_{\ell'}\frac{dP^m_\ell}{dx}\,dx$$
$$+ C^{\mathrm{a}}_{\ell,m}C^{\mathrm{b}\,*}_{\ell',m'}m\frac{f^{\mathrm{a}}_{\ell}Z^{\mathrm{b}\,*}_{\ell'}}{kR}\int_{-1}^{1}P^{m}_{\ell}(x)P^{m'}_{\ell'}(x)\,\frac{x}{\sqrt{1-x^2}}\,dx\Bigg\}\ell'(\ell'+1). \tag{3.44}$$

Analytical solutions for the integral equation (3.39) can be calculated using the following recurence relationships

$$\sqrt{1-x^2}P^m_\ell(x) = \frac{1}{2\ell+1}\left[P^{m+1}_{\ell-1}(x) - P^{m+1}_{\ell+1}(x)\right], \tag{3.45}$$

$$\sqrt{1-x^2}P_\ell^m(x) = \frac{1}{2\ell+1}[(\ell-m+1)(\ell-m+2)P_{\ell+1}^{m-1}(x) - (\ell+m-1)(\ell+m)P_{\ell-1}^{m-1}(x)], \tag{3.46}$$

$$\sqrt{1-x^2}P_\ell^{m-1}(x) = \frac{1}{2\ell+1}\left[P_{\ell-1}^m(x) - P_{\ell+1}^m(x)\right], \tag{3.47}$$

$$\sqrt{1-x^2}P_\ell^{m+1}(x) = \frac{1}{2\ell+1}[(\ell-m)(\ell-m+1)P_{\ell+1}^m(x) - (\ell+m)(\ell+m+1)P_{\ell-1}^m(x)], \tag{3.48}$$

so it can be written that

$$\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m+1}(x) = \frac{1}{2\ell'+1}[(\ell'-m)(\ell'-m+1)\,\frac{dP_\ell^m}{dx}P_{\ell'+1}^m(x) - (\ell'+m)(\ell'+m+1)\,\frac{dP_\ell^m}{dx}P_{\ell'-1}^m(x)], \tag{3.49}$$

$$\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m-1}(x) = \frac{1}{2\ell'+1}\left[\frac{dP_\ell^m}{dx}P_{\ell'-1}^m(x) - \frac{dP_\ell^m}{dx}P_{\ell'+1}^m(x)\right], \tag{3.50}$$

where it has been considered that the sum over ℓ is performed first, and then one performs the sum over ℓ' because that is how the code has been implemented. This ordering is arbitrary and could be reversed in another implementation without affecting the final result. Thus, it is always satisfied that $\ell \geq \ell'$, which helps to simplify the expressions.

3.1.1 Case $m' = m+1$ with positive m

$$\int_{-1}^{1}\frac{P_\ell^m P_{\ell'}^{m+1}}{\sqrt{1-x^2}}dx = 0, \tag{3.51}$$

$$\int_{-1}^{1}\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m+1}(x) = -2\frac{(\ell+1)(\ell+m)!}{(2\ell+1)(\ell-m-1)!}\delta_{\ell,\ell'}, \tag{3.52}$$

$$\int_{-1}^{1}x\frac{P_\ell^m P_{\ell'}^{m+1}}{\sqrt{1-x^2}}dx = -2\frac{(\ell+m)!}{(2\ell+1)(\ell-m-1)!}\delta_{\ell,\ell'}, \tag{3.53}$$

$$\int_{-1}^{1}x\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m+1}(x) = -2\delta_{\ell'+1,\ell}\frac{(\ell+1)(\ell+m)!}{(2\ell+1)(2\ell'+1)(\ell'-m-1)!}. \tag{3.54}$$

3.1.2 Case $m' = m+1$ with negative m

$$\begin{aligned}
\int_{-1}^{1}\frac{P_\ell^{-|m|}P_{\ell'}^{-|m|+1}}{\sqrt{1-x^2}}dx &= \int_{-1}^{1}\frac{P_\ell^{-|m|}P_{\ell'}^{-(|m|-1)}}{\sqrt{1-x^2}}dx \\
&= \int_{-1}^{1}(-1)^|m|\frac{(l-|m|)!}{(l+|m|)!}P_\ell^{|m|}(-1)^{|m|-1}\frac{(l'-m+1)!}{(l'+m-1)!}P_{\ell'}^{|m|-1}\frac{1}{\sqrt{1-x^2}}dx \\
&= -\frac{(l-|m|)!}{(l+|m|)!}\frac{(l'-|m|+1)!}{(l'+|m|-1)!}\int_{-1}^{1}P_\ell^{|m|}P_{\ell'}^{|m|-1}\frac{1}{\sqrt{1-x^2}}dx
\end{aligned}$$

$$= 2\frac{(l-|m|)!}{(l+|m|)!}\delta_{\ell',\ell-2n+1} = 2\frac{(l+m)!}{(l-m)!}\delta_{\ell',\ell-2n+1}, \tag{3.55}$$

if we define $\alpha_{\ell,m}^{\ell',m'} = -\Big[(l-m)!\Big/(l+m)!\Big]\Big[(l'-m+1)!\Big/(l'+m-1)!\Big]$

$$\int_{-1}^{1}\sqrt{1-x^2}\,\frac{dP_\ell^{-|m|}}{dx}P_{\ell'}^{-|m|+1}(x) = \alpha_{\ell,|m|}^{\ell',|m|'}\int_{-1}^{1}\sqrt{1-x^2}\,\frac{dP_\ell^{|m|}}{dx}P_{\ell'}^{|m|-1}(x)$$
$$= 2m\frac{(l-|m|)!}{(l+|m|)!}\delta_{\ell',\ell-2n-2} - \frac{2l}{2l+1}\frac{(l-|m|+1)!}{(l+|m|)!}\delta_{\ell',\ell}, \tag{3.56}$$
$$\int_{-1}^{1}x\frac{P_\ell^{-|m|}P_{\ell'}^{-|m|+1}}{\sqrt{1-x^2}}dx = 2\frac{(l-|m|)!}{(l+|m|)!}\delta_{\ell,\ell'+2n+2} + \frac{2}{2\ell+1}\frac{(\ell-|m|+1)!}{(l+|m|)!}\delta_{\ell,\ell'}, \tag{3.57}$$
$$\int_{-1}^{1}x\sqrt{1-x^2}\,\frac{dP_\ell^{-|m|}}{dx}P_{\ell'}^{-|m|+1}(x) = -\frac{1}{2\ell'+1}\frac{(\ell-|m|)!}{(\ell+|m|)!}\Big\{\Big[(\ell'-|m|)(\ell'-|m|+1)-$$
$$-(\ell'+|m|+1)(\ell'+|m|)\Big]\delta_{\ell,\ell'+2n_1+1} + \Big[(\ell-|m|)(\ell'-|m|) + \frac{(\ell+|m|)(\ell+|m|-1)}{(2\ell+1)}\Big]\delta_{\ell,\ell'+1}\Big\}, \tag{3.58}$$

where n could take any value in $\mathbb{N}$ for which $\ell \geq \ell' \geq 0$.

3.1.3 Case $m' = m-1$ with positive m

$$\int_{-1}^{1}\frac{P_\ell^m P_{\ell'}^{m-1}}{\sqrt{1-x^2}}dx = \int_{-1}^{1}\frac{P_\ell^{m'+1}P_{\ell'}^{m'}}{\sqrt{1-x^2}}dx = -2\frac{(\ell'+m')!}{(\ell'-m')!}\delta_{\ell',\ell-2n-1}, \tag{3.59}$$
$$\int_{-1}^{1}\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m-1}(x) = \frac{(\ell'-1+m)!}{(\ell'+1-m)!}\Big[\frac{2\ell'}{2\ell'+1}(\ell'-m+1)\delta_{\ell,\ell'}$$
$$-2m\delta_{\ell,\ell'+2n+2}\Big], \tag{3.60}$$
$$\int_{-1}^{1}x\frac{P_\ell^m P_{\ell'}^{m-1}}{\sqrt{1-x^2}}dx = -\frac{2(\ell'+m-1)!}{(2\ell'+1)(\ell'-m)!}\delta_{\ell,\ell'} - \frac{2(\ell'+m-1)!}{(\ell'-m+1)!}\delta_{\ell,\ell'+2n+2}, \tag{3.61}$$
$$\int_{-1}^{1}x\sqrt{1-x^2}\,\frac{dP_\ell^m}{dx}P_{\ell'}^{m-1}(x) = \frac{(\ell'+m-1)!}{(\ell'-m+1)!}\Big\{-2m\delta_{\ell',\ell-1-2n_1}$$
$$\frac{1}{2\ell'+1}\left[(\ell'-m)(\ell-m) + \frac{(\ell+1)(\ell'+1)}{2\ell+1}\right]\delta_{\ell,\ell'+1}\Big\}, \tag{3.62}$$

where n could take any value in $\mathbb{N}$ for which $\ell \geq \ell' \geq 0$.

3.1.4 Case $m' = m-1$ with negative m

$$\int_{-1}^{1}\frac{P_\ell^{-|m|}P_{\ell'}^{-|m|-1}}{\sqrt{1-x^2}}dx = \int_{-1}^{1}\frac{P_\ell^{-|m|}P_{\ell'}^{-(|m|+1)}}{\sqrt{1-x^2}}dx \tag{3.63}$$
$$= \int_{-1}^{1}(-1)^|m|\frac{(l-|m|)!}{(l+|m|)!}P_\ell^{|m|}(-1)^{|m|+1}\frac{(l'-|m|-1)!}{(l'+|m|+1)!}P_{\ell'}^{|m|+1}\frac{1}{\sqrt{1-x^2}}dx \tag{3.64}$$
$$= -\frac{(l-|m|)!}{(l+|m|)!}\frac{(l'-|m|-1)!}{(l'+|m|+1)!}\int_{-1}^{1}P_\ell^{|m|}P_{\ell'}^{|m|+1}\frac{1}{\sqrt{1-x^2}}dx = 0, \tag{3.65}$$

$$\int_{-1}^{1} \sqrt{1-x^2}\,\frac{dP_\ell^{|m|}}{dx} P_{\ell'}^{|m|-1}(x) = 2\frac{(\ell-|m|)!}{(\ell+|m|+1)!}\frac{(\ell+1)}{(2\ell+1)}\delta_{\ell,\ell'}, \tag{3.66}$$

$$\int_{-1}^{1} x\frac{P_\ell^{|m|}P_{\ell'}^{|m|-1}}{\sqrt{1-x^2}}dx = \frac{(\ell-|m|)!}{(\ell+|m|+1)!}\frac{2}{2\ell+1}\delta_{\ell,\ell'}, \tag{3.67}$$

$$\int_{-1}^{1} x\sqrt{1-x^2}\,\frac{dP_\ell^{-|m|}}{dx} P_{\ell'}^{-|m|-1}(x) = 2\frac{(\ell-|m|)!}{(\ell+|m|)!}\frac{\ell+1}{(2\ell+1)(2\ell'+1)}\delta_{\ell,\ell'+1}. \tag{3.68}$$

These results provide a robust foundation for understanding the angular momentum transfer from swift electrons to spherical nanoparticles, forming the basis for further investigations into several material properties, as well as their potential applications in electron tweezers.

4

Progress in the Computational Case Analysis: AMT to Aluminum and Gold Nanoparticles

4.1 Convergence Analysis

The goal of this Section is to develop the framework for evaluating the angular momentum transfer to nanoparticles calculating the integral of the spectral densities in the frequency domain, which can often be highly oscillatory or extend over infinite domains. When analytical methods become insufficient, numerical integration techniques, such as quadrature methods, provide a robust framework for approximating these integrals.

This Section delves into the application of various quadrature techniques. We explore several well-known quadrature rules, emphasizing Gaussian quadrature due to its high precision, and further enhance it with Gauss-Kronrod quadrature for error estimation. These methods are not only crucial for obtaining accurate results but also for controlling the numerical error. Additionally, the Section discusses advanced integration techniques, such as double exponential quadrature, to manage the challenges of improper integrals.

Finally, we address the issue of multipolar truncation in spectral density calculations, where an infinite series must be approximated by a finite sum. Through careful convergence testing, we establish the criteria for selecting a truncation point that ensures numerical precision without unnecessary computational overhead.

4.2 Numerical Integration Using Quadratures

Numerical integration is an essential tool when analytical solutions of integrals are not available. In such cases, quadrature methods provide a reliable way to approximate the value of definite integrals by discretizing the integral into a weighted sum of function evaluations at specific points [38–40].

Consider an integral of the form

$$I = \int_a^b f(x)\,dx, \tag{4.1}$$

where $f(x)$ is a continuous function over the interval $[a, b]$. Quadrature methods approximate this integral as

$$I \approx \sum_{i=1}^{N} w_i f(x_i), \tag{4.2}$$

where x_i are the quadrature points (also known as nodes), and w_i are the associated weights. The choice of the quadrature points and weights depends on the specific quadrature rule used. Common methods include trapezoidal, Simpson's rule, and Gaussian quadrature.

Gaussian quadrature is particularly powerful because it maximizes the degree of accuracy by selecting both the quadrature points and the weights optimally for polynomials of a given degree. For an N-point Gaussian quadrature, the integral is approximated by evaluating $f(x)$ at N specific points within the interval, yielding Eq. (4.2) where x_i and w_i are chosen to provide an exact result for polynomials of degree $2N - 1$ [38–40].

The accuracy of the quadrature method depends on both the choice of the quadrature rule and the number of points N used in the approximation. Increasing N generally improves accuracy but at the cost of computational complexity. Therefore, error estimation and the optimal choice of N are crucial, particularly in cases involving highly oscillatory functions.

Quadrature methods are particularly useful in the context of this work, where integrals of the spectral densities in the frequency domain are evaluated numerically to determine the angular momentum transfer to nanoparticles. By applying appropriate quadrature techniques, we ensure that these integrals are computed with the necessary accuracy for our theoretical and computational analysis.

4.3 Gaussian Quadrature and Error Estimation

Gaussian quadrature on its own does not provide an intrinsic measure of error, which makes it difficult to determine whether the chosen number of points N is sufficient for the desired accuracy. To overcome this limitation, we employ the Gauss-Kronrod quadrature rule, which not only improves the accuracy of the approximation but also enables an effective error estimation [38–40].

The Gauss-Kronrod quadrature is an extension of Gaussian quadrature that adds extra points to the Gaussian nodes, effectively creating a higher-order quadrature rule that provides both an accurate integral approximation and an error estimation. Specifically, for an N-point Gaussian quadrature, the Gauss-Kronrod rule uses $2N + 1$ points, but the Gaussian points are a subset of the Kronrod points. This allows for simultaneous evaluation using both quadrature rules:

$$I_{\text{Gauss}} = \sum_{i=1}^{N} w_i^{\text{G}} f(x_i), \tag{4.3}$$

and

$$I_{\text{Kronrod}} = \sum_{i=1}^{2N+1} w_i^{\text{K}} f(x_i), \tag{4.4}$$

where w_i^{G} and w_i^{K} are the weights for the Gaussian and Gauss-Kronrod quadrature rules, respectively.

The key advantage of this method is that the difference between I_{Gauss} and I_{Kronrod} provides an estimate of the error in the Gaussian quadrature approximation. The error estimate ϵ can be expressed as

$$\epsilon = |I_{\mathrm{Kronrod}} - I_{\mathrm{Gauss}}|, \tag{4.5}$$

allowing us to assess the reliability of the computed integral and adapt the number of quadrature points accordingly.

Due to the need for precise integration in the computation of spectral densities and other physical quantities in this work, it is critical to control and minimize numerical errors. The Gauss-Kronrod quadrature offers effective means to achieve this by providing both a highly accurate approximation of the integral and a built-in error estimate. This allows us to dynamically adjust the number of quadrature points until the desired accuracy is achieved.

In the context of integrating functions with rapidly varying behavior or extended spectral ranges, the error estimation provided by the Gauss-Kronrod rule ensures that we maintain control over the numerical precision. This method is particularly useful when handling oscillatory integrands or integrals over large domains, which are common in the evaluation of angular momentum transfer for nanoparticles.

For these reasons, we will adopt Gauss-Kronrod quadrature in this work as the primary method for numerical integration. Its error estimation capabilities ensure that the results are both accurate and computationally efficient, making it the optimal choice for the complex integrals encountered in this work.

4.3.1 Choosing a Cutoff Frequency: Approximating Improper Integrals as Definite Integrals

The main throwback of the Gaussian Quadrature is that it can not be applied to improper integrals as the ones required to calculate the AMT, as the integral in frequencies encompasses an infinite domains. An improper integral is of the form:

$$I = \int_a^\infty f(\omega)\, d\omega. \tag{4.6}$$

As standard numerical techniques, such as Gaussian quadrature, are only valid for integrals over finite intervals, it is necessary to introduce a cutoff frequency ω_{cut} to handle this limitation, effectively transforming the improper integral into a definite one:

$$I = \int_a^{\omega_{\mathrm{cut}}} f(\omega)\, d\omega + \int_{\omega_{\mathrm{cut}}}^\infty f(\omega)\, d\omega \approx \int_a^{\omega_{\mathrm{cut}}} f(\omega)\, d\omega. \tag{4.7}$$

The key challenge is selecting a cutoff frequency ω_{cut} such that the contribution from the interval $[\omega_{\mathrm{cut}}, \infty)$ is negligible or can be estimated with sufficient accuracy. This ensures that the integral can be accurately approximated without compromising numerical precision.

The Fig. 4.1 illustrates the two distinct regions of one of the real integral kernels developed in this work: S_1, where Gauss-Kronrod quadrature is used to evaluate the integral from the lower bound a to ω_{cut}, and S_2, where the Double Exponential [38–40] (Exp-Sinh) quadrature is applied to estimate the tail contribution from ω_{cut} to infinity.

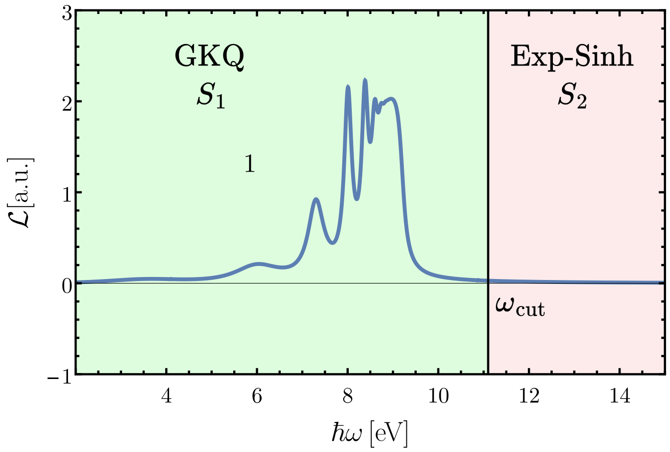

Figure 4.1: Schematic representation of the numerical integration process. The section S_1 (green area) corresponds to the finite integral from a to ω_{cut}, evaluated using Gauss-Kronrod quadrature. The section S_2 (red area) corresponds to the tail integral from ω_{cut} to infinity, approximated using Exp-Sinh quadrature. The cutoff frequency ω_{cut} is chosen such that the integrand $f(\omega)$ is sufficiently small in the S_2 region, ensuring a controlled and negligible error contribution.

4.3.2 Double Exponential Quadrature for Improper Integrals

To accurately account for the contribution of the integral from ω_{cut} to infinity, we employ the double exponential quadrature method, also known as the tanh-sinh or exp-sinh quadrature. This technique is particularly effective for approximating integrals over semi-infinite and infinite intervals. Consider an improper integral of the form:

$$I_\infty = \int_{\omega_{\text{cut}}}^{\infty} f(\omega)\, d\omega, \tag{4.8}$$

where the upper bound extends to infinity. The double exponential quadrature transforms the infinite integration domain into a finite one by a change of variables that forces the integrand to decay exponentially as ω approaches infinity. This transformation makes the integral manageable for standard numerical methods.

The key advantage of the double exponential method is that it ensures the integrand decays rapidly for large ω, allowing us to approximate the tail of the integral with a high degree of accuracy. By combining Gauss-Kronrod quadrature over the finite interval $[0, \omega_{\text{cut}}]$ and double exponential quadrature for the interval $[\omega_{\text{cut}}, \infty)$, the entire integral can be computed with precision while maintaining a controlled error threshold.

This two-step integration strategy is illustrated schematically in Fig. 4.1. In the figure, the section labeled S_1 (green area) represents the finite interval over which Gauss-Kronrod quadrature is applied, while the section labeled S_2 (red area) corresponds to the tail region, where the Exp-Sinh quadrature is employed to capture the contribution from ω_{cut} to infinity. The choice of the cutoff frequency ω_{cut} ensures that the integrand $f(\omega)$ is sufficiently small beyond this point, thereby minimizing any potential error from truncating the domain.

The selection of the cutoff frequency ω_{cut} is critical to ensure that the integral over $[\omega_{\text{cut}}, \infty)$ is small enough to neglect or it can be integrated with an acceptable level of error. The criterion for choosing ω_{cut} is that the integrand $f(\omega)$ should decay sufficiently fast beyond this frequency. Practically, this means that we numerically evaluate the integral for increasing values of ω_{cut} until the improper integral falls below a predefined error threshold, ϵ:

$$\left| \int_{\omega_{\text{cut}}}^{\infty} f(\omega)\, d\omega \right| < \epsilon. \tag{4.9}$$

Once this condition is satisfied, we can confidently truncate the improper integral at ω_{cut} and approximate it as a definite integral.

By using the double exponential quadrature for the tail contribution and selecting an appropriate cutoff frequency, we ensure that the error introduced by truncating the infinite domain is minimized. This approach provides a reliable means to approximate improper integrals in the context of angular momentum transfer and similar calculations, where high precision is essential for obtaining accurate physical results.

4.3.3 Multipolar Truncation: Approximating an Infinite Series with a Finite Sum

The computation of the spectral density involves an infinite sum of multipolar terms. However, for practical computational purposes, it is necessary to impose a truncation point, ℓ_{cut}, at which the multipolar summation is finished. This allows the infinite sum to be approximated as:

$$\sum_{1}^{\infty} = \sum_{1}^{\ell_{\text{cut}}} + \sum_{\ell_{\text{cut}}}^{\infty} \approx \sum_{1}^{\ell_{\text{cut}}}. \tag{4.10}$$

The challenge lies in selecting ℓ_{cut} such that it encompasses all the dominant multipolar contributions. To ensure this, we conducted a numerical comparison between the sum evaluated up to ℓ_{cut} and the sum extended to $\ell_{\text{cut}} + 1$. Convergence is achieved when the difference between these two sums falls below a predefined threshold, indicating that the inclusion of additional terms would not significantly alter the result. This threshold represents the acceptable level of numerical error for the calculation.

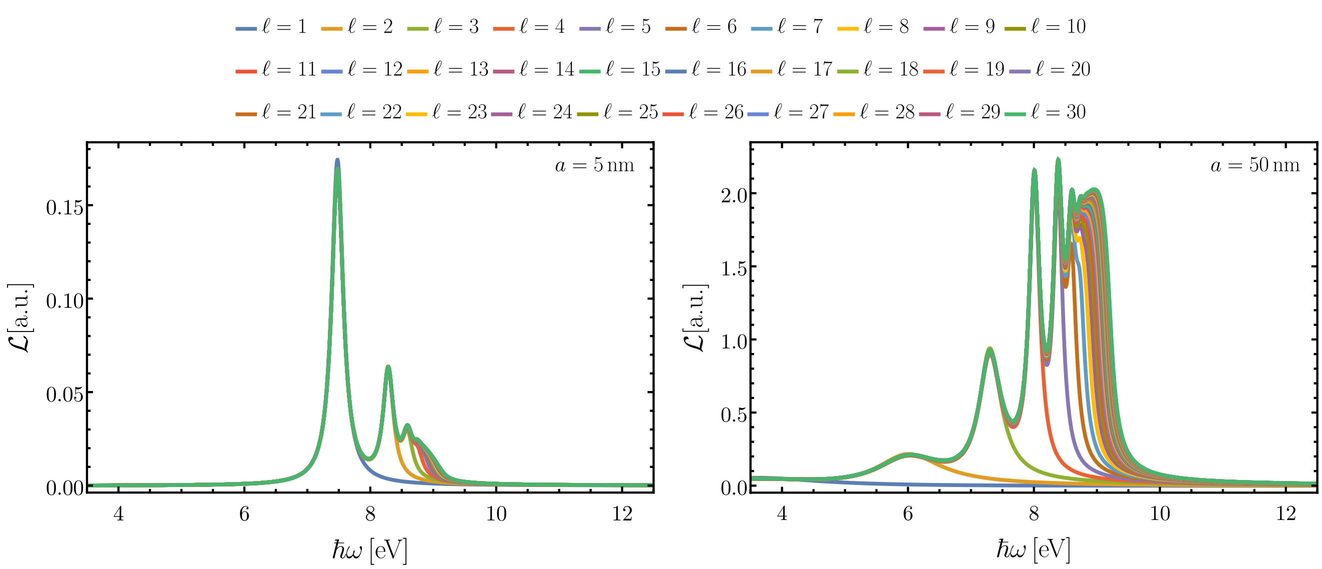

Figure 4.2: Convergence of the spectral density as higher-order multipolar terms are added. The sum stabilizes once the cut-off ℓ_{cut} sufficiently captures the dominant contributions.

Figure 4.2 graphically illustrates the process of progressively adding higher multipolar terms until convergence in the spectral density is attained. The plot demonstrates how the spectral density stabilizes as more terms are included, confirming that the chosen ℓ_{cut} is sufficient to capture all relevant contributions.

The convergence analysis techniques outlined in this Section provide the foundation for accurate and reliable computational results in the evaluation of angular momentum transfer from swift electrons to nanoparticles. In the following Section, we apply these methods to a detailed numerical convergence analysis, assessing the performance of our quadrature-based approach in calculating angular momentum transfer for various nanoparticle configurations. This analysis not only validates the accuracy of our numerical methods but also provides insights into their efficiency and applicability to complex physical systems.

4.4 Angular Momentum Transfer from a Swift Electron to Spherical Nanoparticles: Drude-like (Aluminum) and Lorentzian-based (Gold) Dielectric Responses

Understanding the angular momentum transfer (AMT) requires analyzing the spectral density described by Eq. (3.39). For small nanoparticles (on the order of ~ 1 nm), it has been reported that the dipolar contribution is sufficient to describe the angular momentum transfer. In such cases, the dipole approximation provides a good estimate for calculating AMT [28, 29, 41]. However, as the nanoparticle size increases, higher-order multipolar contributions become significant, as demonstrated in Fig. 4.3. Additionally, for this Drude-like aluminum nanoparticle, each multipolar resonance exhibits a redshift as the radius increases.

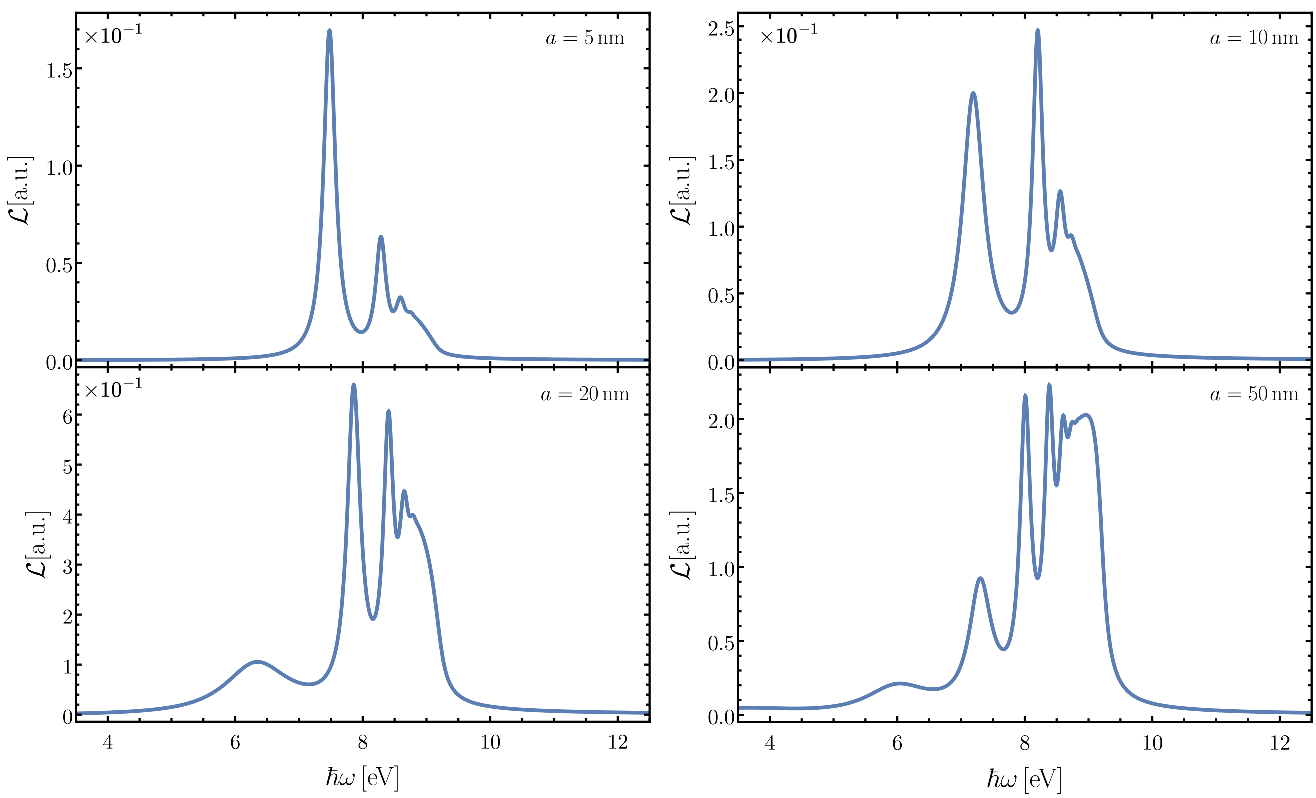

Figure 4.3: Spectral densities, in atomic units (a.u.), as a function of excitation energy for nanoparticles with radii of 5 nm, 10 nm, 20 nm, and 50 nm. As the nanoparticle size increases, higher-order multipolar resonances become more relevant and shift to higher excitation energies. The interaction is modeled considering an electron with energy of 200 keV ($v \sim 0.7c$) and corresponding impact parameters of $b = 5.5$, 11, 21, and 51 nm, respectively.

As the nanoparticle radius increases, the magnitude of the spectral density also increases, as shown in Fig. 4.3. It is reasonable to infer that the integral of the spectral density, representing the total angular momentum transfer, increases with nanoparticle size. However, for a fair comparison, Fig. 4.4 presents the AMT normalized by the nanoparticle's radius cubed ($a^3 = 3V/4\pi$), which is related to the volume of the nanoparticle. This figure indicates that smaller nanoparticles retain angular momentum more efficiently.

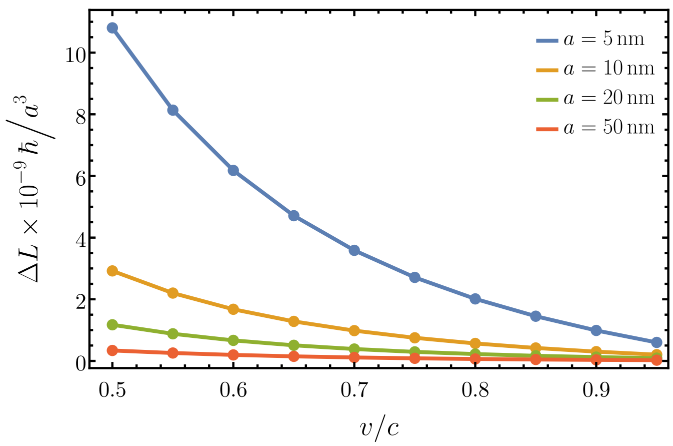

Figure 4.4: Angular momentum transfer to Al NPs normalized by $a^3 \sim$ Volume, as a function of the swift electron speed (which is related to the electron beam energy). The interaction is modeled considering an impact parameters of $b = 5.5$, 11, 21, and 51 nm, respectively.

To explore a more realistic dielectric response, we model Gold nanoparticles (NPs) using a sum of several Lorentzian functions [42]. This approach allows us to generalize the behavior observed in the simpler Drude model for metallic responses. The resulting spectral structure becomes significantly richer and more complex, covering a broader range of the spectrum with multiple resonances. These resonances reflect the complex interactions between the incident electron and the electronic excitations in the nanoparticle.

As the radius of the nanoparticle increases, the spectral response may appear similar at first glance, but a closer analysis reveals a difference: the spectral density decreases more rapidly for larger particles, indicating a less efficient interaction at higher excitation energies. This behavior can be observed in Fig. 4.5, where the spectral density is plotted for different nanoparticle radii. Larger particles exhibit sharper drops in spectral density beyond certain energy thresholds.

Additionally, a comparative study of the angular momentum transfer (AMT), normalized by the volume of the nanoparticle (using the $a^3 \propto V$ scaling factor), shows a consistent trend with the findings from the Drude-modeled Aluminum case. Despite the richer spectral features, the general conclusion holds: smaller nanoparticles tend to transfer angular momentum more efficiently due to their stronger absorption properties over a given volume.

As the radius of the Gold nanoparticle increases, there is a corresponding rise in the spectral density, as illustrated in Fig. 4.5. This suggests that the total angular momentum transfer grows with larger nanoparticle sizes. Again, to make a meaningful comparison, Fig. 4.6 presents the angular momentum transfer normalized by the nanoparticle's volume, represented as the radius cubed ($a^3 = 3V/4\pi$). This normalization highlights that smaller Gold nanoparticles demonstrate a more efficient transfer of angular momentum. These findings are consistent with the results previously observed for Aluminum nanoparticles, where similar trends were noted.

By comparing Figures 4.4 and 4.6, it can be observed that the AMT to Au nanoparticles is an order

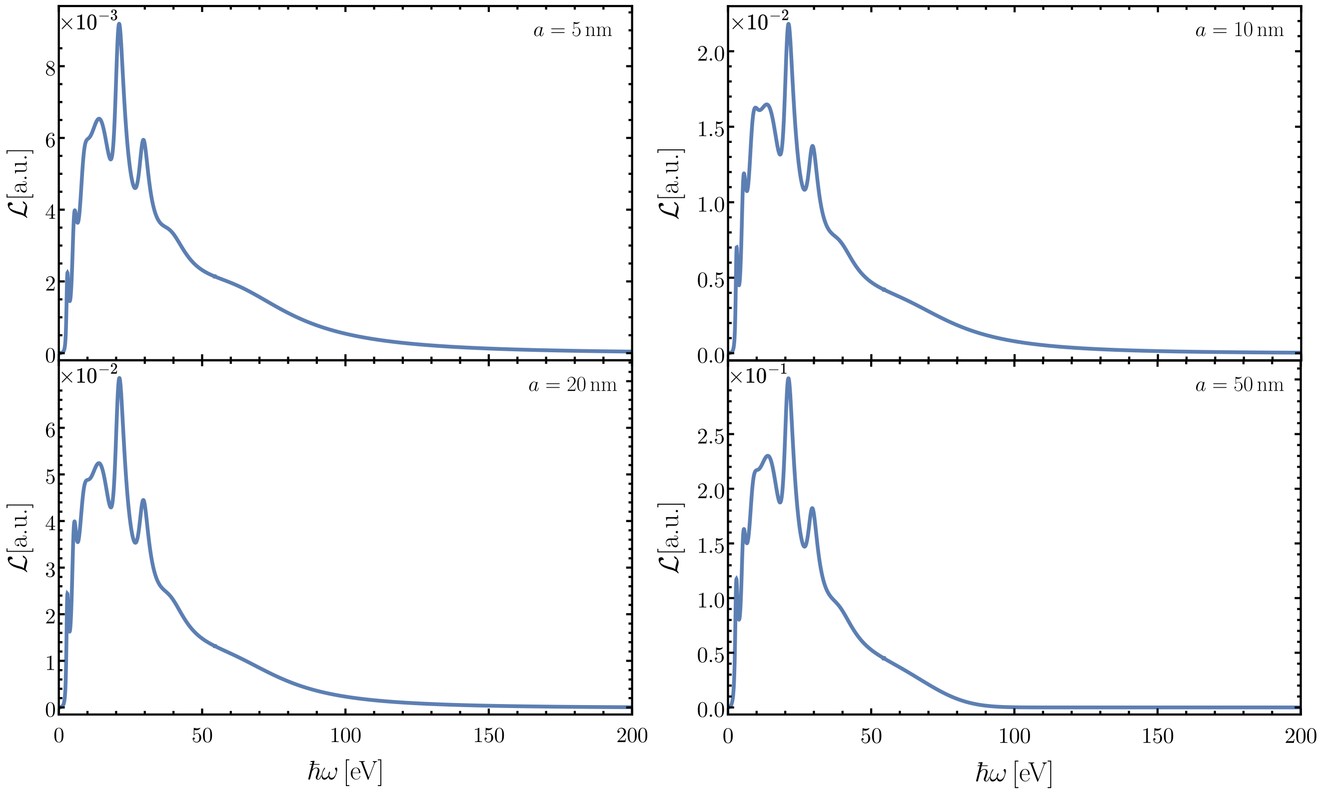

Figure 4.5: Spectral densities, in atomic units (a.u.), as a function of excitation energy for gold nanoparticles with varying radii. The plot demonstrates how spectral density decays more rapidly for larger particles, highlighting the diminishing strength of resonances as particle size increases. The interaction is modeled considering an electron with energy of 200 keV ($v \sim 0.7c$) and corresponding impact parameters of $b = 5.5$, 11, 21, and 51 nm, respectively.

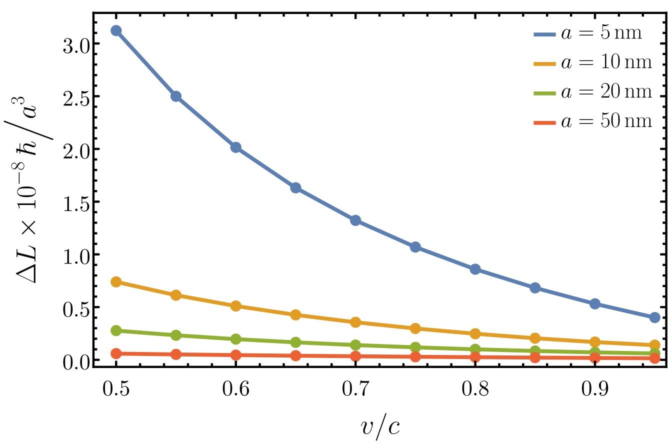

Figure 4.6: Angular momentum transfer to Au NPs normalized by $a^3 \sim$ Volume, as a function of the swift electron speed (which is related to the electron beam energy). The interaction is modeled considering an impact parameters of $b = 5.5$, 11, 21, and 51 nm, respectively.

of magnitude greater than that to Al nanoparticles. This difference arises from the broader distribution of resonances in Au NPs, where the lower-energy resonances exhibit higher intensity. These resonances coincide with regions of stronger electromagnetic fields, leading to a more pronounced AMT. This highlights the significant role of material properties and resonance behavior in determining the efficiency of angular momentum transfer.

5 Work plan (proposed activities to complete the project)

5.1 Research Progress

In summary, the progress achieved during the first year of my doctoral studies includes the development of analytical solutions for the spectral densities of angular momentum transfer from swift electrons to spherical nanoparticles. Additionally, I have written and implemented a numerical code to compute the frequency integrals of the spectral density using quadrature methods, allowing for an efficient calculation of the angular momentum transfer.

During this period, we also submitted a peer-reviewed article on the angular momentum transfer from swift electrons to small non-spherical nanoparticles within the dipole approximation. Details of this publication can be found in the List of Publications Section.

5.2 Research Timeline

This Section outlines the planned timeline for my research project, focusing on completing my current investigations on angular momentum transfer and extending the study to broader areas, such as interactions between fast electrons and nanoparticles. My research will span both theoretical and computational studies, and aims to contribute further to the understanding of nanoparticle interactions.

- **August – December 2024**: Complete the convergence tests and finalize the development of the code that calculates angular momentum transfer to spherical nanoparticles. These tests are critical to ensure the accuracy and reliability of the computational results, and will form the foundation of the final phase of this research.

- **January – June 2025**: Carry out calculations and convergence analysis for both plasmonic and dielectric materials to quantify angular momentum transfer. During this period, I will also begin writing a manuscript to document the results of my doctoral research, with a view to submitting it for a peer reviewed publication.

- **July – December 2025**: Finalize and submit the paper containing the primary results from my doctoral research for publication. This will be the culmination of my work on angular momentum transfer in spherical nanoparticle systems with dielectric responses.
- **July 2025 – June 2027**: Pursue further research on the interaction of fast electrons with nanoparticles, potentially including a PhD research internship abroad. This phase will focus on exploring several advanced topics, such as:
  - Investigating the impact of different nanoparticle geometries, with potential applications to chiral studies.
  - Studying the effects of the nonlocal dielectric function on electron beams interacting near or inside nanoparticles.
  - Exploring radiation reaction effects in nanoparticles caused by electron beam interactions.
  - Investigating electron interactions with non-plasmonic materials (dielectric or magnetic) in the context of angular momentum transfer, linear momentum transfer, and electron energy loss spectroscopy (EELS).

List of publications

During my doctoral studies, I contributed to the publication of the following peer-reviewed works:

1.- Angular momentum transfer from swift electron to small non-spherical nanoparticles in the dipolar approximation (2024)

J. L. Briseño-Gómez, A. López-Tercero, J. Á. Castellanos-Reyes, A. Reyes-Coronado, Ultramicroscopy, Volume 264, 114005 (2024).

In electron microscopy and material science, the interaction between swift electrons and nanostructures plays a crucial role. In this work, we present a comprehensive theoretical investigation into the influence of nanoparticle geometry on angular momentum transfer from a swift electron to small symmetric nanoparticles. Specifically, we focus on spheroidal and polyhedral nanoparticles, analyzing their response within the framework of the dipolar approximation. By exploring various geometric configurations, we showed how subtle changes in shape affect the efficiency and dynamics of angular momentum transfer. This study not only enhances our understanding of the underlying physical mechanisms but also provides valuable insights for potential applications, such as nanomanipulation or fabrication of nanomaterials.

We are also preparing two manuscripts for submission as peer-reviewed publications:

- The first manuscript, authored by **J. L. Briseño-Gómez**, A. Reyes-Coronado, and M. Á. Pizaña-Juárez, focuses on a unique academic and educational-focused problem solution. It involves calculating the electromotive force hypothetically generated on the Moon due to its natural motion through the surrounding spatial magnetic field. This study explores theoretical aspects of electromagnetism in a novel context and aims to provide a pedagogical tool for teaching complex physical concepts and numerical methods.
- The second manuscript, co-authored by **J. L. Briseño-Gómez**, J. Castrejón-Figueroa, E. E. Viveros-Armas, and A. Reyes-Coronado, addresses the efficient calculation of linear momentum transfer from swift electrons to spherical nanoparticles. This work spans the entire nanoscale and examines both plasmonic and dielectric materials using a causal dielectric function, providing a

comprehensive study on electron-nanoparticle interactions that could have implications for advanced nanotechnology applications.

- The main manuscript, co-authored by **J. L. Briseño-Gómez** and A. Reyes-Coronado, focuses on the efficient calculation of angular momentum transfer from swift electrons to spherical nanoparticles. The preliminary results presented in this manuscript are drawn from the work outlined in this PhD Candidate Protocol.

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